

Mathematics

1,000 Digital SAT Math Questions

Practice items matched to the published Digital SAT math blueprint.

Algebra	350 questions, about 35%
Advanced Math	350 questions, about 35%
Problem-Solving and Data Analysis	150 questions, about 15%
Geometry and Trigonometry	150 questions, about 15%

Of these 1,000 items, 799 are four-option multiple choice and 201 are student-produced responses, the kind you type as a number. On the Digital SAT itself, about 75% are multiple choice and about 25% are student-produced responses. Each item is marked Easy, Medium, or Hard so the bank covers the range from an easier module through a harder second module.

These are original practice items, not College Board questions, and this is not an official SAT test. A calculator is appropriate throughout, as on the Digital SAT. For student-produced responses, enter an integer or a fraction in lowest terms. Do not use a mixed number. Geometry items use the standard reference facts: triangle angle sum 180° , a circle has 360° , $c = 2\pi r$, $A = \pi r^2$, rectangle area ℓw , triangle area $\frac{1}{2}bh$, Pythagorean theorem, 30° - 60° - 90° and 45° - 45° - 90° side ratios, cylinder volume $\pi r^2 h$, and right-triangle trigonometry.

The answer key begins after question 1000.

Algebra

1. Linear equations in one variable · Easy · MC

What is the solution to the equation $5x + 11 = 51$?

- A. 8
- B. 9
- C. 40
- D. 7

2. Linear equations in one variable · Easy · MC

What is the solution to the equation $2x + 10 = 14$?

- A. 2
- B. 3
- C. 1
- D. 4

3. Linear equations in one variable · Easy · SPR

What is the solution to the equation $2x + 8 = 12$?

4. Linear equations in one variable · Easy · MC

What is the solution to the equation $3x + 9 = 42$?

- A. 33
- B. 10
- C. 12
- D. 11

5. Linear equations in one variable · Easy · SPR

What is the solution to the equation $2x + 15 = 31$?

6. Linear equations in one variable · Easy · MC

What is the solution to the equation $3x + 9 = 15$?

- A. 3
- B. 2
- C. 1
- D. 6

7. Linear equations in one variable · Easy · MC

What is the solution to the equation $6x + 3 = 39$?

- A. 5
- B. 36
- C. 7
- D. 6

8. Linear equations in one variable · Easy · MC

What is the solution to the equation $5x + 11 = 71$?

- A. 13
- B. 12
- C. 60
- D. 11

9. Linear equations in one variable · Easy · MC

What is the solution to the equation $6x + 15 = 57$?

- A. 7
- B. 8
- C. 42
- D. 6

10. Linear equations in one variable · Easy · MC

What is the solution to the equation $4x + 2 = 18$?

- A. 4
- B. 5
- C. 16
- D. 3

11. Linear equations in one variable · Easy · MC

What is the solution to the equation $5x + 14 = 59$?

- A. 45
- B. 9
- C. 10
- D. 8

12. Linear equations in one variable · Easy · SPR

What is the solution to the equation $6x + 5 = 41$?

13. Linear equations in one variable · Easy · MC

What is the solution to the equation $2x + 3 = 15$?

- A. 5
- B. 6
- C. 7
- D. 12

14. Linear equations in one variable · Easy · MC

What is the solution to the equation $2x + 2 = 14$?

- A. 7
- B. 12
- C. 6
- D. 5

15. Linear equations in one variable · Easy · SPR

What is the solution to the equation $8x + 9 = 49$?

16. Linear equations in one variable · Easy · MC

What is the solution to the equation $8x + 7 = 31$?

- A. 3
- B. 4
- C. 2
- D. 24

17. Linear equations in one variable · Easy · SPR

What is the solution to the equation $3x + 6 = 24$?

18. Linear equations in one variable · Easy · MC

What is the solution to the equation $6x + 6 = 78$?

- A. 13
- B. 72
- C. 12
- D. 11

19. Linear equations in one variable · Easy · MC

What is the solution to the equation $4x + 15 = 27$?

- A. 2
- B. 3
- C. 12
- D. 4

20. Linear equations in one variable · Easy · SPR

What is the solution to the equation $5x + 2 = 17$?

21. Linear equations in one variable · Easy · SPR

What is the solution to the equation $4x + 13 = 53$?

22. Linear equations in one variable · Easy · MC

What is the solution to the equation $5x + 11 = 56$?

- A. 10
- B. 9
- C. 45
- D. 8

23. Linear equations in one variable · Easy · SPR

What is the solution to the equation $5x + 8 = 23$?

24. Linear equations in one variable · Easy · SPR

What is the solution to the equation $3x + 1 = 22$?

25. Linear equations in one variable · Easy · MC

What is the solution to the equation $4x + 2 = 46$?

- A. 12
- B. 10
- C. 44
- D. 11

26. Linear equations in one variable · Easy · MC

What is the solution to the equation $6x + 3 = 45$?

- A. 7
- B. 8
- C. 6
- D. 42

27. Linear equations in one variable · Easy · MC

What is the solution to the equation $3x + 13 = 37$?

- A. 24
- B. 7
- C. 8
- D. 9

28. Linear equations in one variable · Easy · SPR

What is the solution to the equation $5x + 10 = 65$?

29. Linear equations in one variable · Easy · MC

What is the solution to the equation $5x + 9 = 49$?

- A. 40
- B. 8
- C. 7
- D. 9

30. Linear equations in one variable · Easy · SPR

What is the solution to the equation $8x + 8 = 32$?

31. Linear equations in one variable · Easy · MC

What is the solution to the equation $5x + 14 = 49$?

- A. 7
- B. 6
- C. 35
- D. 8

32. Linear equations in one variable · Easy · MC

What is the solution to the equation $3x + 2 = 32$?

- A. 10
- B. 11
- C. 9
- D. 30

33. Linear equations in one variable · Easy · MC

What is the solution to the equation $2x + 13 = 33$?

- A. 11
- B. 9
- C. 10
- D. 20

34. Linear equations in one variable · Easy · MC

What is the solution to the equation $6x + 1 = 25$?

- A. 24
- B. 5
- C. 3
- D. 4

35. Linear equations in one variable · Easy · MC

What is the solution to the equation $5x + 9 = 64$?

- A. 12
- B. 11
- C. 55
- D. 10

36. Linear equations in one variable · Easy · MC

What is the solution to the equation $3x + 3 = 36$?

- A. 11
- B. 12
- C. 10
- D. 33

37. Linear equations in one variable · Easy · MC

What is the solution to the equation $3x + 2 = 17$?

- A. 5
- B. 15
- C. 6
- D. 4

38. Linear equations in one variable · Easy · MC

What is the solution to the equation $3x + 15 = 39$?

- A. 24
- B. 9
- C. 8
- D. 7

39. Linear equations in one variable · Easy · MC

What is the solution to the equation $6x + 12 = 66$?

- A. 8
- B. 54
- C. 10
- D. 9

40. Linear equations in one variable · Easy · MC

What is the solution to the equation $2x + 7 = 29$?

- A. 22
- B. 10
- C. 12
- D. 11

41. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $8x + 6 = 6x + 26$?

- A. 20
- B. 16
- C. -10
- D. 10

42. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $7x - 4 = 2x + 16$?

43. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $6x - 1 = x + 54$?

- A. 12
- B. 11
- C. -11
- D. 55

44. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $4x - 5 = 3x - 8$?

- A. -3
- B. 0
- C. 3
- D. -2

45. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $4x + 8 = 2x + 26$?

46. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $5x - 1 = 2x + 20$?

- A. 7
- B. -7
- C. 21
- D. 9

47. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $5x - 2 = x + 30$?

48. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $4x = 3x + 4$?

49. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $7x + 10 = 5x + 20$?

50. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $4x + 1 = 2x + 21$?

- A. 20
- B. 12
- C. -10
- D. 10

51. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $8x + 7 = 5x + 28$?

52. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $8x - 3 = 4x - 15$?

- A. 1
- B. -3
- C. 3
- D. -12

53. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $7x = 5x - 8$?

54. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $5x + 10 = 4x + 17$?

55. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $7x + 10 = 2x - 15$?

- A. 5
- B. -25
- C. -3
- D. -5

56. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $3x - 1 = x - 7$?

- A. -6
- B. -3
- C. -2
- D. 3

57. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $3x - 8 = 2x - 13$?

- A. -3
- B. -5
- C. 5
- D. -4

58. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $8x - 5 = x + 2$?

- A. 1
- B. -1
- C. 7
- D. 2

59. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $5x - 6 = x - 22$?

60. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $4x - 3 = 2x - 9$?

61. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $7x + 5 = 4x - 7$?

- A. -4
- B. 0
- C. 4
- D. -12

62. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $4x + 2 = 3x + 3$?

- A. -1
- B. 1
- C. 4
- D. 2

63. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $4x - 6 = 3x + 4$?

64. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $3x + 1 = 2x + 5$?

- A. 5
- B. 6
- C. -4
- D. 4

65. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $8x + 10 = 5x + 16$?

- A. -2
- B. 7
- C. 2
- D. 6

66. Linear equations in one variable · Medium · SPR

What value of x satisfies the equation $4x + 10 = 3x + 9$?

67. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $8x + 9 = 7x + 12$?

- A. 3
- B. -3
- C. 10
- D. 4

68. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $7x = 5x + 18$?

- A. 18
- B. 9
- C. -9
- D. 14

69. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $9x - 6 = 7x + 10$?

- A. -8
- B. 15
- C. 16
- D. 8

70. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $4x + 5 = x + 26$?

- A. 7
- B. 21
- C. 8
- D. -7

71. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $4x + 5 = 2x + 29$?

- A. -12
- B. 12
- C. 24
- D. 14

72. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $8x - 7 = 4x - 23$?

- A. 4
- B. -16
- C. 0
- D. -4

73. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $9x + 1 = 5x + 49$?

- A. 12
- B. 48
- C. -12
- D. 17

74. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $8x + 3 = 4x + 27$?

- A. 6
- B. 10
- C. -6
- D. 24

75. Linear equations in one variable · Medium · MC

What value of x satisfies the equation $3x - 3 = x - 1$?

- A. -2
- B. 1
- C. 2
- D. -1

76. Linear equations in one variable · Hard · MC

What is the solution to $3(x + 4) = 3x + 12$?

- A. -9
- B. 9
- C. 10
- D. 12

77. Linear equations in one variable · Hard · MC

What is the solution to $3(x + 3) = 6x - 9$?

- A. 4
- B. 6
- C. -6
- D. 9

78. Linear equations in one variable · Hard · MC

What is the solution to $3(x + 5) = 6x + 6$?

- A. $-\frac{1}{3}$
- B. -3
- C. 3
- D. 6

79. Linear equations in one variable · Hard · MC

What is the solution to $4(x + 4) = 3x + 13$?

- A. 3
- B. 9
- C. 1
- D. -3

80. Linear equations in one variable · Hard · MC

What is the solution to $4(x + 1) = 2x$?

- A. -2
- B. -7
- C. $-\frac{1}{2}$
- D. 2

81. Linear equations in one variable · Hard · MC

What is the solution to $4(x + 2) = 3x + 5$?

- A. -10
- B. 1
- C. -3
- D. 3

82. Linear equations in one variable · Hard · MC

What is the solution to $5(x + 2) = 5x + 10$?

- A. -3
- B. 8
- C. 3
- D. 4

83. Linear equations in one variable · Hard · SPR

What is the solution to $3(x + 6) = 6x + 27$?

84. Linear equations in one variable · Hard · MC

What is the solution to $2(x + 4) = 6x - 24$?

- A. 8
- B. -8
- C. 10
- D. 7

85. Linear equations in one variable · Hard · SPR

What is the solution to $5(x + 6) = 6x + 27$?

86. Linear equations in one variable · Hard · MC

What is the solution to $2(x + 2) = 3x + 1$?

- A. -3
- B. 3
- C. 1
- D. 5

87. Linear equations in one variable · Hard · MC

What is the solution to $4(x + 1) = 5x + 1$?

- A. 0
- B. 7
- C. 3
- D. -3

88. Linear equations in one variable · Hard · MC

What is the solution to $4(x + 3) = 5x + 9$?

- A. 7
- B. -6
- C. -3
- D. 3

89. Linear equations in one variable · Hard · MC

What is the solution to $2(x + 6) = 6x - 20$?

- A. 10
- B. -8
- C. $\frac{13}{2}$
- D. 8

90. Linear equations in one variable · Hard · MC

What is the solution to $5(x + 6) = 3x + 32$?

- A. -1
- B. 1
- C. 6
- D. 13

91. Linear equations in one variable · Hard · MC

What is the solution to $5(x + 2) = 4x + 16$?

- A. -6
- B. 11
- C. 6
- D. 14

92. Linear equations in one variable · Hard · MC

What is the solution to $4(x + 3) = 5x + 4$?

- A. 12
- B. 8
- C. -8
- D. -1

93. Linear equations in one variable · Hard · SPR

What is the solution to $5(x + 1) = 5x + 5$?

94. Linear equations in one variable · Hard · MC

What is the solution to $4(x + 3) = 2x + 22$?

- A. $\frac{19}{2}$
- B. 9
- C. 5
- D. -5

95. Linear equations in one variable · Hard · SPR

What is the solution to $3(x + 1) = 2x + 11$?

96. Linear equations in one variable · Hard · MC

What is the solution to $4(x + 1) = 2x + 8$?

- A. $\frac{7}{2}$
- B. 6
- C. 2
- D. -2

97. Linear equations in one variable · Hard · MC

What is the solution to $4(x + 3) = 6x + 4$?

- A. $-\frac{1}{2}$
- B. 4
- C. -4
- D. 8

98. Linear equations in one variable · Hard · MC

What is the solution to $4(x + 1) = 6x - 14$?

- A. $\frac{15}{2}$
- B. -9
- C. 13
- D. 9

99. Linear equations in one variable · Hard · MC

What is the solution to $2(x + 1) = 4x - 14$?

- A. 8
- B. 10
- C. $\frac{15}{2}$
- D. -8

100. Linear equations in one variable · Hard · SPR

What is the solution to $2(x + 4) = 5x + 20$?

101. Number of solutions · Medium · MC

How many solutions does the equation $2x + 3 = 2x + 8$ have?

- A. Exactly two solutions
- B. No solution
- C. Infinitely many solutions
- D. Exactly one solution

102. Number of solutions · Medium · MC

How many solutions does the equation $5(6x + 6) = 30x + 30$ have?

- A. Infinitely many solutions
- B. Exactly one solution
- C. Exactly two solutions
- D. No solution

103. Number of solutions · Medium · MC

How many solutions does the equation $2(3x + 7) = 6x + 14$ have?

- A. Exactly one solution
- B. Exactly two solutions
- C. No solution
- D. Infinitely many solutions

104. Number of solutions · Medium · MC

How many solutions does the equation $2(5x + 2) = 10x + 4$ have?

- A. Exactly two solutions
- B. No solution
- C. Infinitely many solutions
- D. Exactly one solution

105. Number of solutions · Medium · MC

How many solutions does the equation $4(5x + 5) = 20x + 20$ have?

- A. Exactly one solution
- B. Exactly two solutions
- C. No solution
- D. Infinitely many solutions

106. Number of solutions · Medium · MC

How many solutions does the equation $7x + 7 = 7x + 10$ have?

- A. No solution
- B. Infinitely many solutions
- C. Exactly one solution
- D. Exactly two solutions

107. Number of solutions · Medium · MC

How many solutions does the equation $3(5x + 2) = 15x + 6$ have?

- A. Exactly one solution
- B. Exactly two solutions
- C. No solution
- D. Infinitely many solutions

108. Number of solutions · Medium · MC

How many solutions does the equation $4x + 7 = 4x + 13$ have?

- A. Exactly one solution
- B. Exactly two solutions
- C. No solution
- D. Infinitely many solutions

109. Number of solutions · Medium · MC

How many solutions does the equation $3x + 3 = 3x + 9$ have?

- A. No solution
- B. Infinitely many solutions
- C. Exactly one solution
- D. Exactly two solutions

110. Number of solutions · Medium · MC

How many solutions does the equation $4(8x + 2) = 32x + 8$ have?

- A. Infinitely many solutions
- B. Exactly one solution
- C. Exactly two solutions
- D. No solution

111. Number of solutions · Medium · MC

How many solutions does the equation $4x + 8 = 4x + 9$ have?

- A. Exactly two solutions
- B. No solution
- C. Infinitely many solutions
- D. Exactly one solution

112. Number of solutions · Medium · MC

How many solutions does the equation $2(5x + 7) = 10x + 14$ have?

- A. Exactly one solution
- B. Exactly two solutions
- C. No solution
- D. Infinitely many solutions

113. Number of solutions · Hard · MC

How many solutions does the equation $5x + 1 = 5x + 5$ have?

- A. No solution
- B. Infinitely many solutions
- C. Exactly one solution
- D. Exactly two solutions

114. Number of solutions · Hard · MC

How many solutions does the equation $4(3x + 4) = 12x + 16$ have?

- A. Infinitely many solutions
- B. Exactly one solution
- C. Exactly two solutions
- D. No solution

115. Number of solutions · Hard · MC

How many solutions does the equation $6x + 7 = 6x + 12$ have?

- A. Exactly two solutions
- B. No solution
- C. Infinitely many solutions
- D. Exactly one solution

116. Number of solutions · Hard · MC

How many solutions does the equation $5(3x + 1) = 15x + 5$ have?

- A. No solution
- B. Infinitely many solutions
- C. Exactly one solution
- D. Exactly two solutions

117. Number of solutions · Hard · MC

How many solutions does the equation $4x + 4 = 4x + 7$ have?

- A. Infinitely many solutions
- B. Exactly one solution
- C. Exactly two solutions
- D. No solution

118. Number of solutions · Hard · MC

How many solutions does the equation $4x + 6 = 4x + 12$ have?

- A. No solution
- B. Infinitely many solutions
- C. Exactly one solution
- D. Exactly two solutions

119. Number of solutions · Hard · MC

How many solutions does the equation $2(7x + 8) = 14x + 16$ have?

- A. Infinitely many solutions
- B. Exactly one solution
- C. Exactly two solutions
- D. No solution

120. Number of solutions · Hard · MC

How many solutions does the equation $2(5x + 4) = 10x + 8$ have?

- A. Infinitely many solutions
- B. Exactly one solution
- C. Exactly two solutions
- D. No solution

121. Linear inequalities · Easy · MC

What is the solution set of $5x - 5 < 45$?

- A. $x < 9$
- B. $x \leq 10$
- C. $x < 10$
- D. $x > 10$

122. Linear inequalities · Easy · MC

What is the solution set of $2x - 3 < 15$?

- A. $x < 8$
- B. $x \leq 9$
- C. $x < 9$
- D. $x > 9$

123. Linear inequalities · Easy · MC

What is the solution set of $4x - 5 < 7$?

- A. $x < 2$
- B. $x \leq 3$
- C. $x < 3$
- D. $x > 3$

124. Linear inequalities · Easy · MC

What is the solution set of $4x - 7 < 25$?

- A. $x \leq 8$
- B. $x < 8$
- C. $x > 8$
- D. $x < 7$

125. Linear inequalities · Easy · MC

What is the solution set of $3x - 5 < 19$?

- A. $x < 7$
- B. $x \leq 8$
- C. $x < 8$
- D. $x > 8$

126. Linear inequalities · Easy · MC

What is the solution set of $3x - 1 < 20$?

- A. $x \leq 7$
- B. $x < 7$
- C. $x > 7$
- D. $x < 6$

127. Linear inequalities · Easy · MC

What is the solution set of $2x - 5 < 7$?

- A. $x \leq 6$
- B. $x < 6$
- C. $x > 6$
- D. $x < 5$

128. Linear inequalities · Easy · MC

What is the solution set of $3x - 2 < 7$?

- A. $x < 2$
- B. $x \leq 3$
- C. $x < 3$
- D. $x > 3$

129. Linear inequalities · Easy · MC

What is the solution set of $5x - 3 < 27$?

- A. $x < 5$
- B. $x \leq 6$
- C. $x < 6$
- D. $x > 6$

130. Linear inequalities · Easy · MC

What is the solution set of $2x - 4 < 4$?

- A. $x < 3$
- B. $x \leq 4$
- C. $x < 4$
- D. $x > 4$

131. Linear inequalities · Easy · MC

What is the solution set of $2x - 4 < 14$?

- A. $x > 9$
- B. $x < 8$
- C. $x \leq 9$
- D. $x < 9$

132. Linear inequalities · Easy · MC

What is the solution set of $2x - 5 < 5$?

- A. $x > 5$
- B. $x < 4$
- C. $x \leq 5$
- D. $x < 5$

133. Linear inequalities · Medium · MC

What is the solution set of $5x - 7 < 18$?

- A. $x < 5$
- B. $x > 5$
- C. $x < 4$
- D. $x \leq 5$

134. Linear inequalities · Medium · MC

What is the solution set of $2x - 6 < 2$?

- A. $x < 4$
- B. $x > 4$
- C. $x < 3$
- D. $x \leq 4$

135. Linear inequalities · Medium · MC

What is the solution set of $2x - 4 < 12$?

- A. $x > 8$
- B. $x < 7$
- C. $x \leq 8$
- D. $x < 8$

136. Linear inequalities · Medium · MC

What is the solution set of $2x - 8 < 12$?

- A. $x < 9$
- B. $x \leq 10$
- C. $x < 10$
- D. $x > 10$

137. Linear inequalities · Medium · MC

What is the solution set of $4x - 3 < 17$?

- A. $x > 5$
- B. $x < 4$
- C. $x \leq 5$
- D. $x < 5$

138. Linear inequalities · Medium · MC

What is the solution set of $2x - 6 < 4$?

- A. $x < 5$
- B. $x > 5$
- C. $x < 4$
- D. $x \leq 5$

139. Linear inequalities · Medium · MC

What is the solution set of $5x - 2 < 33$?

- A. $x > 7$
- B. $x < 6$
- C. $x \leq 7$
- D. $x < 7$

140. Linear inequalities · Medium · MC

What is the solution set of $5x - 1 < 44$?

- A. $x \leq 9$
- B. $x < 9$
- C. $x > 9$
- D. $x < 8$

141. Linear inequalities · Medium · MC

What is the solution set of $4x - 5 < 31$?

- A. $x \leq 9$
- B. $x < 9$
- C. $x > 9$
- D. $x < 8$

142. Linear inequalities · Medium · MC

What is the solution set of $3x - 8 < -2$?

- A. $x > 2$
- B. $x < 1$
- C. $x \leq 2$
- D. $x < 2$

143. Linear inequalities · Hard · MC

What is the solution set of $-4x + 8 < -4$?

- A. $x > 3$
- B. $x < 3$
- C. $x > 4$
- D. $x \geq 3$

144. Linear inequalities · Hard · MC

What is the solution set of $-2x + 1 < -17$?

- A. $x > 10$
- B. $x \geq 9$
- C. $x > 9$
- D. $x < 9$

145. Linear inequalities · Hard · MC

What is the solution set of $-3x + 2 < -25$?

- A. $x > 9$
- B. $x < 9$
- C. $x > 10$
- D. $x \geq 9$

146. Linear inequalities · Hard · MC

What is the solution set of $-3x + 3 < -12$?

- A. $x > 5$
- B. $x < 5$
- C. $x > 6$
- D. $x \geq 5$

147. Linear inequalities · Hard · MC

What is the solution set of $-5x + 4 < -31$?

- A. $x < 7$
- B. $x > 8$
- C. $x \geq 7$
- D. $x > 7$

148. Linear inequalities · Hard · MC

What is the solution set of $-5x + 5 < -25$?

- A. $x < 6$
- B. $x > 7$
- C. $x \geq 6$
- D. $x > 6$

149. Linear inequalities · Hard · MC

What is the solution set of $-4x + 1 < -11$?

- A. $x > 4$
- B. $x \geq 3$
- C. $x > 3$
- D. $x < 3$

150. Linear inequalities · Hard · MC

What is the solution set of $-2x + 8 < -8$?

- A. $x > 8$
- B. $x < 8$
- C. $x > 9$
- D. $x \geq 8$

151. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $2x + 3y > 14$?

- A. $(0, 0)$
- B. $(5, 2)$
- C. $(2, 2)$
- D. $(4, 0)$

152. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $2x + 1y > 8$?

- A. $(0, 0)$
- B. $(4, 2)$
- C. $(1, 2)$
- D. $(3, 0)$

153. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $3x + 1y > 12$?

- A. $(3, 6)$
- B. $(0, 6)$
- C. $(2, 4)$
- D. $(0, 0)$

154. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $1x + 2y > 3$?

- A. $(-1, 1)$
- B. $(1, -1)$
- C. $(0, 0)$
- D. $(2, 1)$

155. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $1x + 2y > 6$?

- A. $(4, -1)$
- B. $(0, 0)$
- C. $(5, 1)$
- D. $(2, 1)$

156. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $2x + 1y > 13$?

- A. $(4, 1)$
- B. $(6, -1)$
- C. $(0, 0)$
- D. $(7, 1)$

157. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $2x + 1y > 6$?

- A. $(2, 4)$
- B. $(-1, 4)$
- C. $(1, 2)$
- D. $(0, 0)$

158. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $2x + 1y > 7$?

- A. $(2, 1)$
- B. $(0, 0)$
- C. $(3, 3)$
- D. $(0, 3)$

159. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $1x + 2y > 11$?

- A. $(0, 0)$
- B. $(2, 5)$
- C. $(-1, 5)$
- D. $(1, 3)$

160. Linear inequalities · Medium · MC

Which point (x, y) is a solution of the inequality $2x + 1y > 9$?

- A. $(3, 1)$
- B. $(0, 0)$
- C. $(4, 3)$
- D. $(1, 3)$

161. Linear equations in two variables · Easy · MC

What is the slope of the line through $(3, 3)$ and $(7, 5)$?

- A. 4
- B. $\frac{1}{2}$
- C. $-\frac{1}{2}$
- D. 2

162. Linear equations in two variables · Easy · SPR

What is the slope of the line through (2, 1) and (3, 4)?

163. Linear equations in two variables · Easy · MC

What is the slope of the line through (3, 3) and (4, 7)?

A. $\frac{1}{4}$

B. 4

C. 1

D. -4

164. Linear equations in two variables · Easy · SPR

What is the slope of the line through (0, 2) and (2, 8)?

165. Linear equations in two variables · Easy · MC

What is the slope of the line through (0, 0) and (1, 6)?

A. 1

B. $\frac{1}{6}$

C. -6

D. 6

166. Linear equations in two variables · Easy · SPR

What is the slope of the line through (2, 4) and (4, 5)?

167. Linear equations in two variables · Easy · MC

What is the slope of the line through (3, 2) and (4, 3)?

A. 1

B. 0

C. 2

D. -1

168. Linear equations in two variables · Easy · MC

What is the slope of the line through (3, 2) and (5, 6)?

A. 2

B. $\frac{1}{2}$

C. 4

D. -2

169. Linear equations in two variables · Easy · MC

What is the slope of the line through (3, 0) and (5, 4)?

A. $\frac{1}{2}$

B. -2

C. 4

D. 2

170. Linear equations in two variables · Easy · MC

What is the slope of the line through (1, 4) and (4, 7)?

A. 3

B. -1

C. 1

D. 2

171. Linear equations in two variables · Easy · SPR

What is the slope of the line through $(4, 3)$ and $(5, 6)$?

172. Linear equations in two variables · Easy · MC

What is the slope of the line through $(1, 3)$ and $(4, 6)$?

A. -1

B. 2

C. 3

D. 1

173. Linear equations in two variables · Easy · MC

What is the slope of the line through $(0, 0)$ and $(1, 3)$?

A. 1

B. $\frac{1}{3}$

C. 3

D. -3

174. Linear equations in two variables · Easy · SPR

What is the slope of the line through $(3, 0)$ and $(6, 2)$?

175. Linear equations in two variables · Easy · MC

What is the slope of the line through $(2, 2)$ and $(5, 6)$?

A. $\frac{3}{4}$

B. $\frac{4}{3}$

C. 4

D. $-\frac{4}{3}$

176. Linear equations in two variables · Easy · MC

What is the slope of the line through $(0, 0)$ and $(4, 2)$?

A. 4

B. $\frac{1}{2}$

C. $-\frac{1}{2}$

D. 2

177. Linear equations in two variables · Easy · MC

What is the slope of the line through $(3, 0)$ and $(7, 4)$?

A. -1

B. 4

C. 1

D. 2

178. Linear equations in two variables · Easy · SPR

What is the slope of the line through $(1, 1)$ and $(2, 4)$?

179. Linear equations in two variables · Easy · SPR

What is the slope of the line through $(4, 1)$ and $(6, 3)$?

180. Linear equations in two variables · Easy · MC

What is the slope of the line through $(0, 3)$ and $(2, 4)$?

- A. 1
- B. 2
- C. $\frac{1}{2}$
- D. $-\frac{1}{2}$

181. Linear equations in two variables · Easy · MC

What is the slope of the line through $(4, 0)$ and $(6, 1)$?

- A. $-\frac{1}{2}$
- B. $\frac{1}{2}$
- C. 2
- D. 1

182. Linear equations in two variables · Easy · SPR

What is the slope of the line through $(1, 0)$ and $(3, 6)$?

183. Linear functions · Medium · MC

The line with equation $y = 2x + 3$ passes through the point $(3, k)$. What is the value of k ?

- A. 6
- B. 9
- C. 3
- D. 5

184. Linear functions · Medium · MC

The line with equation $y = 4x - 1$ passes through the point $(1, k)$. What is the value of k ?

- A. 3
- B. 0
- C. 5
- D. -3

185. Linear functions · Medium · MC

The line with equation $y = -\frac{1}{2}x + 6$ passes through the point $(2, k)$. What is the value of k ?

- A. -7
- B. 8
- C. 5
- D. $\frac{11}{2}$

186. Linear functions · Medium · MC

The line with equation $y = 2x - 4$ passes through the point $(-1, k)$. What is the value of k ?

- A. 2
- B. -6
- C. -5
- D. -2

187. Linear functions · Medium · MC

The line with equation $y = 2x + 2$ passes through the point $(2, k)$. What is the value of k ?

- A. 6
- B. -6
- C. 2
- D. 4

188. Linear functions · Medium · SPR

The line with equation $y = 4x$ passes through the point $(4, k)$. What is the value of k ?

189. Linear functions · Medium · MC

The line with equation $y = 3x + 5$ passes through the point $(0, k)$. What is the value of k ?

- A. -5
- B. 8
- C. 5
- D. 6

190. Linear functions · Medium · MC

The line with equation $y = -x + 4$ passes through the point $(5, k)$. What is the value of k ?

- A. -1
- B. -9
- C. 3
- D. 9

191. Linear functions · Medium · MC

The line with equation $y = 2x$ passes through the point $(4, k)$. What is the value of k ?

- A. -8
- B. 2
- C. 8
- D. 4

192. Linear functions · Medium · MC

The line with equation $y = 2x - 4$ passes through the point $(4, k)$. What is the value of k ?

- A. -2
- B. 12
- C. 0
- D. 4

193. Linear functions · Medium · MC

The line with equation $y = x - 1$ passes through the point $(-2, k)$. What is the value of k ?

- A. 0
- B. -1
- C. 3
- D. -3

194. Linear functions · Medium · MC

The line with equation $y = -3x + 4$ passes through the point $(3, k)$. What is the value of k ?

- A. 1
- B. -13
- C. -5
- D. 7

195. Linear functions · Medium · MC

The line with equation $y = 4x + 4$ passes through the point $(0, k)$. What is the value of k ?

- A. -4
- B. 5
- C. 8
- D. 4

196. Linear functions · Medium · SPR

The line with equation $y = 3x + 3$ passes through the point $(2, k)$. What is the value of k ?

197. Linear functions · Medium · MC

The line with equation $y = -2x + 1$ passes through the point $(4, k)$. What is the value of k ?

- A. -9
- B. 5
- C. -7
- D. -1

198. Linear functions · Medium · SPR

The line with equation $y = x + 3$ passes through the point $(5, k)$. What is the value of k ?

199. Linear functions · Medium · MC

The line with equation $y = \frac{3}{2}x - 1$ passes through the point $(-2, k)$. What is the value of k ?

- A. -2
- B. $\frac{1}{2}$
- C. -3
- D. -4

200. Linear functions · Medium · SPR

The line with equation $y = 2x + 6$ passes through the point $(-1, k)$. What is the value of k ?

201. Linear functions · Medium · MC

The line with equation $y = 2x + 3$ passes through the point $(5, k)$. What is the value of k ?

- A. 5
- B. 13
- C. 7
- D. 8

202. Linear functions · Medium · MC

The line with equation $y = x + 2$ passes through the point $(-2, k)$. What is the value of k ?

- A. 1
- B. 3
- C. -4
- D. 0

203. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{3}{2}x + 3$. What is the slope of the perpendicular line?

- A. $\frac{3}{2}$
- B. $-\frac{2}{3}$
- C. $-\frac{3}{2}$
- D. $\frac{2}{3}$

204. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{2}{3}x - 3$. What is the slope of the perpendicular line?

- A. $-\frac{3}{2}$
- B. $\frac{2}{3}$
- C. $-\frac{2}{3}$
- D. $\frac{3}{2}$

205. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = x - 2$. What is the slope of the perpendicular line?

- A. 2
- B. 0
- C. 1
- D. -1

206. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = 4x + 3$. What is the slope of the perpendicular line?

- A. -4
- B. 4
- C. $-\frac{1}{4}$
- D. $\frac{1}{4}$

207. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{3}{2}x - 1$. What is the slope of the perpendicular line?

- A. $\frac{3}{2}$
- B. $-\frac{3}{2}$
- C. $\frac{2}{3}$
- D. $-\frac{2}{3}$

208. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{1}{2}x + 1$. What is the slope of the perpendicular line?

- A. $\frac{1}{2}$
- B. 2
- C. $-\frac{1}{2}$
- D. -2

209. Linear equations in two variables · Hard · SPR

A line is perpendicular to the line $y = 3x + 2$. What is the slope of the perpendicular line?

210. Linear equations in two variables · Hard · SPR

A line is perpendicular to the line $y = 2x - 1$. What is the slope of the perpendicular line?

211. Linear equations in two variables · Hard · SPR

A line is perpendicular to the line $y = \frac{1}{3}x + 4$. What is the slope of the perpendicular line?

212. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = x$. What is the slope of the perpendicular line?

- A. 2
- B. -1
- C. 0
- D. 1

213. Linear equations in two variables · Hard · SPR

A line is perpendicular to the line $y = 3x$. What is the slope of the perpendicular line?

214. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{3}{2}x + 4$. What is the slope of the perpendicular line?

- A. $-\frac{2}{3}$
- B. $\frac{3}{2}$
- C. $\frac{2}{3}$
- D. $-\frac{3}{2}$

215. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = 3x + 1$. What is the slope of the perpendicular line?

- A. -3
- B. $\frac{1}{3}$
- C. 3
- D. $-\frac{1}{3}$

216. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = 2x - 3$. What is the slope of the perpendicular line?

- A. -2
- B. $\frac{1}{2}$
- C. $-\frac{1}{2}$
- D. 2

217. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{4}{3}x - 3$. What is the slope of the perpendicular line?

- A. $\frac{3}{4}$
- B. $-\frac{4}{3}$
- C. $\frac{4}{3}$
- D. $-\frac{3}{4}$

218. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = 4x - 2$. What is the slope of the perpendicular line?

- A. $-\frac{1}{4}$
- B. -4
- C. $\frac{1}{4}$
- D. 4

219. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = \frac{1}{2}x + 3$. What is the slope of the perpendicular line?

- A. 2
- B. $-\frac{1}{2}$
- C. -2
- D. $\frac{1}{2}$

220. Linear equations in two variables · Hard · MC

A line is perpendicular to the line $y = 2x + 3$. What is the slope of the perpendicular line?

- A. $\frac{1}{2}$
- B. $-\frac{1}{2}$
- C. -2
- D. 2

221. Linear functions · Easy · MC

A phone battery is at 200% and loses 12 percentage points each hour during a call. Which equation gives p , the amount remaining after h hours, assuming the rate stays constant?

- A. $p = 12h - 200$
- B. $p = 200 + 12h$
- C. $p = 12 - 200h$
- D. $p = 200 - 12h$

222. Linear functions · Easy · MC

A student has \$150 saved and spends \$3 each week. Which equation gives d , the amount remaining after w weeks, assuming the rate stays constant?

- A. $d = 150 + 3w$
- B. $d = 3 - 150w$
- C. $d = 150 - 3w$
- D. $d = 3w - 150$

223. Linear functions · Easy · MC

A tank contains 80 gallons of water and drains at a constant rate of 4 gallons per minute. Which equation gives g , the amount remaining after m minutes, assuming the rate stays constant?

- A. $g = 4m - 80$
- B. $g = 80 + 4m$
- C. $g = 4 - 80m$
- D. $g = 80 - 4m$

224. Linear functions · Easy · MC

A phone battery is at 60% and loses 3 percentage points each hour during a call. Which equation gives p , the amount remaining after h hours, assuming the rate stays constant?

- A. $p = 3 - 60h$
- B. $p = 60 - 3h$
- C. $p = 3h - 60$
- D. $p = 60 + 3h$

225. Linear functions · Easy · MC

A phone battery is at 80% and loses 3 percentage points each hour during a call. Which equation gives p , the amount remaining after h hours, assuming the rate stays constant?

- A. $p = 3h - 80$
- B. $p = 80 + 3h$
- C. $p = 3 - 80h$
- D. $p = 80 - 3h$

226. Linear functions · Easy · MC

A student has \$60 saved and spends \$3 each week. Which equation gives d , the amount remaining after w weeks, assuming the rate stays constant?

- A. $d = 60 + 3w$
- B. $d = 3 - 60w$
- C. $d = 60 - 3w$
- D. $d = 3w - 60$

227. Linear functions · Easy · MC

A phone battery is at 80% and loses 6 percentage points each hour during a call. Which equation gives p , the amount remaining after h hours, assuming the rate stays constant?

- A. $p = 6h - 80$
- B. $p = 80 + 6h$
- C. $p = 6 - 80h$
- D. $p = 80 - 6h$

228. Linear functions · Easy · MC

A tank contains 150 gallons of water and drains at a constant rate of 8 gallons per minute. Which equation gives g , the amount remaining after m minutes, assuming the rate stays constant?

- A. $g = 150 + 8m$
- B. $g = 8 - 150m$
- C. $g = 150 - 8m$
- D. $g = 8m - 150$

229. Linear functions · Easy · MC

A student has \$60 saved and spends \$4 each week. Which equation gives d , the amount remaining after w weeks, assuming the rate stays constant?

- A. $d = 60 - 4w$
- B. $d = 4w - 60$
- C. $d = 60 + 4w$
- D. $d = 4 - 60w$

230. Linear functions · Easy · MC

A student has \$120 saved and spends \$8 each week. Which equation gives d , the amount remaining after w weeks, assuming the rate stays constant?

- A. $d = 8 - 120w$
- B. $d = 120 - 8w$
- C. $d = 8w - 120$
- D. $d = 120 + 8w$

231. Linear functions · Easy · MC

A student has \$60 saved and spends \$6 each week. Which equation gives d , the amount remaining after w weeks, assuming the rate stays constant?

- A. $d = 6 - 60w$
- B. $d = 60 - 6w$
- C. $d = 6w - 60$
- D. $d = 60 + 6w$

232. Linear functions · Easy · MC

A student has \$150 saved and spends \$12 each week. Which equation gives d , the amount remaining after w weeks, assuming the rate stays constant?

- A. $d = 12 - 150w$
- B. $d = 150 - 12w$
- C. $d = 12w - 150$
- D. $d = 150 + 12w$

233. Linear functions · Medium · MC

A student has \$60 saved and spends \$10 each week. Which equation gives d , the amount remaining after w weeks, assuming the rate stays constant?

- A. $d = 10 - 60w$
- B. $d = 60 - 10w$
- C. $d = 10w - 60$
- D. $d = 60 + 10w$

234. Linear functions · Medium · MC

A student has \$120 saved and spends \$12 each week. Which equation gives d , the amount remaining after w weeks, assuming the rate stays constant?

- A. $d = 120 - 12w$
- B. $d = 12w - 120$
- C. $d = 120 + 12w$
- D. $d = 12 - 120w$

235. Linear functions · Medium · MC

A phone battery is at 120% and loses 6 percentage points each hour during a call. Which equation gives p , the amount remaining after h hours, assuming the rate stays constant?

- A. $p = 6 - 120h$
- B. $p = 120 - 6h$
- C. $p = 6h - 120$
- D. $p = 120 + 6h$

236. Linear functions · Medium · MC

A phone battery is at 200% and loses 5 percentage points each hour during a call. Which equation gives p , the amount remaining after h hours, assuming the rate stays constant?

- A. $p = 5 - 200h$
- B. $p = 200 - 5h$
- C. $p = 5h - 200$
- D. $p = 200 + 5h$

237. Linear functions · Medium · MC

A student has \$40 saved and spends \$5 each week. Which equation gives d , the amount remaining after w weeks, assuming the rate stays constant?

- A. $d = 40 - 5w$
- B. $d = 5w - 40$
- C. $d = 40 + 5w$
- D. $d = 5 - 40w$

238. Linear functions · Medium · MC

A tank contains 40 gallons of water and drains at a constant rate of 6 gallons per minute. Which equation gives g , the amount remaining after m minutes, assuming the rate stays constant?

- A. $g = 40 - 6m$
- B. $g = 6m - 40$
- C. $g = 40 + 6m$
- D. $g = 6 - 40m$

239. Linear functions · Medium · MC

A tank contains 200 gallons of water and drains at a constant rate of 4 gallons per minute. Which equation gives g , the amount remaining after m minutes, assuming the rate stays constant?

- A. $g = 4m - 200$
- B. $g = 200 + 4m$
- C. $g = 4 - 200m$
- D. $g = 200 - 4m$

240. Linear functions · Medium · MC

A phone battery is at 120% and loses 8 percentage points each hour during a call. Which equation gives p , the amount remaining after h hours, assuming the rate stays constant?

- A. $p = 120 + 8h$
- B. $p = 8 - 120h$
- C. $p = 120 - 8h$
- D. $p = 8h - 120$

241. Linear functions · Medium · MC

A tank contains 80 gallons of water and drains at a constant rate of 3 gallons per minute. Which equation gives g , the amount remaining after m minutes, assuming the rate stays constant?

- A. $g = 3 - 80m$
- B. $g = 80 - 3m$
- C. $g = 3m - 80$
- D. $g = 80 + 3m$

242. Linear functions · Medium · MC

A tank contains 60 gallons of water and drains at a constant rate of 3 gallons per minute. Which equation gives g , the amount remaining after m minutes, assuming the rate stays constant?

- A. $g = 60 - 3m$
- B. $g = 3m - 60$
- C. $g = 60 + 3m$
- D. $g = 3 - 60m$

243. Linear functions · Medium · MC

A tank contains 240 gallons of water and drains at a constant rate of 8 gallons per minute. Which equation gives g , the amount remaining after m minutes, assuming the rate stays constant?

- A. $g = 8m - 240$
- B. $g = 240 + 8m$
- C. $g = 8 - 240m$
- D. $g = 240 - 8m$

244. Linear functions · Medium · MC

A student has \$60 saved and spends \$8 each week. Which equation gives d , the amount remaining after w weeks, assuming the rate stays constant?

- A. $d = 60 - 8w$
- B. $d = 8w - 60$
- C. $d = 60 + 8w$
- D. $d = 8 - 60w$

245. Linear functions · Medium · MC

A phone battery is at 120% and loses 4 percentage points each hour during a call. Which equation gives p , the amount remaining after h hours, assuming the rate stays constant?

- A. $p = 120 - 4h$
- B. $p = 4h - 120$
- C. $p = 120 + 4h$
- D. $p = 4 - 120h$

246. Systems of linear equations · Easy · SPR

The solution of the system $x + y = 1$ and $x - y = 3$ is (x, y) . What is the value of x ?

247. Systems of linear equations · Easy · SPR

The solution of the system $x + y = -7$ and $x - y = 1$ is (x, y) . What is the value of x ?

248. Systems of linear equations · Easy · MC

The solution of the system $x + y = -5$ and $x - y = -3$ is (x, y) . What is the value of x ?

- A. -3
- B. -4
- C. -5
- D. -1

249. Systems of linear equations · Easy · MC

The solution of the system $x + y = 7$ and $x - y = 3$ is (x, y) . What is the value of x ?

- A. 5
- B. 3
- C. 2
- D. 7

250. Systems of linear equations · Easy · MC

The solution of the system $x + y = 5$ and $x - y = -3$ is (x, y) . What is the value of x ?

- A. 5
- B. 1
- C. -3
- D. 4

251. Systems of linear equations · Easy · MC

The solution of the system $x + y = -6$ and $x - y = 0$ is (x, y) . What is the value of x ?

- A. -6
- B. -2
- C. -3
- D. 0

252. Systems of linear equations · Easy · MC

The solution of the system $x + y = 15$ and $x - y = 1$ is (x, y) . What is the value of x ?

- A. 7
- B. 1
- C. 15
- D. 8

253. Systems of linear equations · Easy · MC

The solution of the system $x + y = 0$ and $x - y = 0$ is (x, y) . What is the value of x ?

- A. 0
- B. 1
- C. -1
- D. 2

254. Systems of linear equations · Easy · MC

The solution of the system $x + y = 3$ and $x - y = -1$ is (x, y) . What is the value of x ?

- A. -1
- B. 2
- C. 1
- D. 3

255. Systems of linear equations · Easy · MC

The solution of the system $x + y = -1$ and $x - y = 3$ is (x, y) . What is the value of x ?

- A. 1
- B. 3
- C. -1
- D. -2

256. Systems of linear equations · Easy · MC

The solution of the system $x + y = 0$ and $x - y = -2$ is (x, y) . What is the value of x ?

- A. 1
- B. -2
- C. -1
- D. 0

257. Systems of linear equations · Easy · MC

The solution of the system $x + y = -5$ and $x - y = 3$ is (x, y) . What is the value of x ?

- A. -5
- B. -1
- C. 3
- D. -4

258. Systems of linear equations · Easy · MC

The solution of the system $x + y = 4$ and $x - y = 2$ is (x, y) . What is the value of x ?

- A. 1
- B. 4
- C. 3
- D. 2

259. Systems of linear equations · Easy · MC

The solution of the system $x + y = -2$ and $x - y = -6$ is (x, y) . What is the value of x ?

- A. -2
- B. 2
- C. -6
- D. -4

260. Systems of linear equations · Easy · MC

The solution of the system $x + y = -4$ and $x - y = 0$ is (x, y) . What is the value of x ?

- A. -4
- B. -1
- C. 0
- D. -2

261. Systems of linear equations · Easy · MC

The solution of the system $x + y = 12$ and $x - y = 0$ is (x, y) . What is the value of x ?

- A. 7
- B. 12
- C. 6
- D. 0

262. Systems of linear equations · Easy · MC

The solution of the system $x + y = 5$ and $x - y = 1$ is (x, y) . What is the value of x ?

- A. 2
- B. 1
- C. 3
- D. 5

263. Systems of linear equations · Easy · MC

The solution of the system $x + y = 11$ and $x - y = 1$ is (x, y) . What is the value of x ?

- A. 1
- B. 6
- C. 11
- D. 5

264. Systems of linear equations · Easy · SPR

The solution of the system $x + y = 2$ and $x - y = -4$ is (x, y) . What is the value of x ?

265. Systems of linear equations · Easy · MC

The solution of the system $x + y = 11$ and $x - y = 5$ is (x, y) . What is the value of x ?

- A. 11
- B. 8
- C. 3
- D. 5

266. Systems of linear equations · Medium · MC

What is the value of x in the solution of $2x + 5y = 36$ and $x + 2y = 15$?

- A. -3
- B. 36
- C. 6
- D. 3

267. Systems of linear equations · Medium · MC

What is the value of x in the solution of $5x + 2y = 6$ and $x - 2y = -18$?

- A. -10
- B. -2
- C. 8
- D. 6

268. Systems of linear equations · Medium · SPR

What is the value of x in the solution of $5x + 2y = 25$ and $x - y = 5$?

269. Systems of linear equations · Medium · MC

What is the value of x in the solution of $3x + 4y = 16$ and $x - y = -11$?

- A. -4
- B. 16
- C. 7
- D. -11

270. Systems of linear equations · Medium · MC

What is the value of x in the solution of $2x + 5y = 51$ and $x - y = 1$?

- A. 7
- B. 8
- C. 1
- D. 51

271. Systems of linear equations · Medium · MC

What is the value of x in the solution of $3x + 2y = -10$ and $x - 2y = -6$?

- A. -5
- B. 1
- C. -10
- D. -4

272. Systems of linear equations · Medium · MC

What is the value of x in the solution of $5x + 2y = 6$ and $x - 2y = 6$?

- A. 2
- B. -2
- C. 6
- D. 4

273. Systems of linear equations · Medium · MC

What is the value of x in the solution of $2x + 3y = 6$ and $2x - y = 6$?

- A. 6
- B. 4
- C. 0
- D. 3

274. Systems of linear equations · Medium · MC

What is the value of x in the solution of $4x + y = -4$ and $x + 2y = 6$?

- A. 4
- B. -2
- C. -4
- D. -6

275. Systems of linear equations · Medium · MC

What is the value of x in the solution of $2x + 3y = 30$ and $x - y = -5$?

- A. 8
- B. 30
- C. -5
- D. 3

276. Systems of linear equations · Medium · SPR

What is the value of x in the solution of $3x + 2y = 32$ and $2x - y = 12$?

277. Systems of linear equations · Medium · SPR

What is the value of x in the solution of $2x + 5y = 54$ and $x - y = -1$?

278. Systems of linear equations · Medium · SPR

What is the value of x in the solution of $3x + 4y = 9$ and $x - 2y = 13$?

279. Systems of linear equations · Medium · MC

What is the value of x in the solution of $2x + 5y = 40$ and $x + 2y = 16$?

- A. 8
- B. -8
- C. 40
- D. 0

280. Systems of linear equations · Medium · MC

What is the value of x in the solution of $2x + 5y = 18$ and $x + 2y = 8$?

- A. 4
- B. 18
- C. 5
- D. 2

281. Systems of linear equations · Medium · MC

What is the value of x in the solution of $2x + 3y = 9$ and $2x - y = 13$?

- A. 7
- B. -1
- C. 6
- D. 9

282. Systems of linear equations · Medium · MC

What is the value of x in the solution of $3x + 4y = 20$ and $x - y = 9$?

- A. 9
- B. 8
- C. -1
- D. 20

283. Systems of linear equations · Medium · MC

What is the value of x in the solution of $4x + y = -8$ and $3x + y = -4$?

- A. -8
- B. -12
- C. -4
- D. 8

284. Systems of linear equations · Medium · SPR

What is the value of x in the solution of $5x + 2y = 19$ and $x - 2y = -13$?

285. Systems of linear equations · Medium · SPR

What is the value of x in the solution of $2x + 5y = 37$ and $x + 2y = 15$?

286. Systems of linear equations · Hard · MC

What is the value of $x + y$ in the solution of $3x + 2y = 29$ and $x + 2y = 15$?

- A. 4
- B. 3
- C. 7
- D. 11

287. Systems of linear equations · Hard · SPR

What is the value of $x + y$ in the solution of $5x + 2y = 11$ and $2x - y = -10$?

288. Systems of linear equations · Hard · MC

What is the value of $x + y$ in the solution of $2x + 3y = 36$ and $x - 2y = -10$?

- A. 14
- B. -2
- C. 6
- D. 8

289. Systems of linear equations · Hard · MC

What is the value of $x + y$ in the solution of $5x + 2y = 9$ and $2x - y = 9$?

- A. 6
- B. 3
- C. 0
- D. -3

290. Systems of linear equations · Hard · MC

What is the value of $x + y$ in the solution of $2x + 5y = 22$ and $x - 2y = 2$?

- A. 8
- B. 6
- C. 2
- D. 4

291. Systems of linear equations · Hard · MC

What is the value of $x + y$ in the solution of $3x + 2y = 21$ and $x + 2y = 11$?

- A. 8
- B. 5
- C. 2
- D. 3

292. Systems of linear equations · Hard · MC

What is the value of $x + y$ in the solution of $5x + 2y = 37$ and $3x + y = 22$?

- A. 7
- B. 1
- C. 8
- D. 6

293. Systems of linear equations · Hard · SPR

What is the value of $x + y$ in the solution of $2x + 3y = 22$ and $x - y = 1$?

294. Systems of linear equations · Hard · MC

What is the value of $x + y$ in the solution of $2x + 3y = 24$ and $x - 2y = -2$?

- A. 10
- B. 2
- C. 4
- D. 6

295. Systems of linear equations · Hard · SPR

What is the value of $x + y$ in the solution of $4x + y = 10$ and $2x - y = -4$?

296. Systems of linear equations · Hard · MC

What is the value of $x + y$ in the solution of $4x + y = 20$ and $3x + y = 14$?

- A. 2
- B. 10
- C. -4
- D. 6

297. Systems of linear equations · Hard · MC

What is the value of $x + y$ in the solution of $2x + 3y = 17$ and $x - 2y = -2$?

- A. 4
- B. 7
- C. 3
- D. 1

298. Systems of linear equations · Hard · MC

What is the value of $x + y$ in the solution of $2x + 5y = -20$ and $x - 2y = 8$?

- A. -4
- B. 4
- C. 0
- D. -20

299. Systems of linear equations · Hard · SPR

What is the value of $x + y$ in the solution of $4x + y = 27$ and $3x + y = 20$?

300. Systems of linear equations · Hard · MC

What is the value of $x + y$ in the solution of $2x + 5y = 13$ and $2x - y = -5$?

- A. 3
- B. -1
- C. 2
- D. -4

301. Systems of linear equations · Medium · MC

How many solutions does the system $y = 4x - 2$ and $y = 4x + 2$ have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. None

302. Systems of linear equations · Medium · MC

How many solutions does the system $y = 5x + 3$ and $3y = 15x + 9$ have?

- A. None
- B. Exactly one
- C. Exactly two
- D. Infinitely many

303. Systems of linear equations · Medium · MC

How many solutions does the system $y = 4x$ and $y = 4x + 2$ have?

- A. Exactly two
- B. Infinitely many
- C. None
- D. Exactly one

304. Systems of linear equations · Medium · MC

How many solutions does the system $y = 2x + 4$ and $2y = 4x + 8$ have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. None

305. Systems of linear equations · Medium · MC

How many solutions does the system $y = 4x + 3$ and $3y = 12x + 9$ have?

- A. Exactly two
- B. Infinitely many
- C. None
- D. Exactly one

306. Systems of linear equations · Medium · MC

How many solutions does the system $y = 3x + 5$ and $y = 3x + 7$ have?

- A. None
- B. Exactly one
- C. Exactly two
- D. Infinitely many

307. Systems of linear equations · Medium · MC

How many solutions does the system $y = 5x + 5$ and $y = 5x + 9$ have?

- A. Exactly two
- B. Infinitely many
- C. None
- D. Exactly one

308. Systems of linear equations · Medium · MC

How many solutions does the system $y = 4x + 4$ and $y = 4x + 8$ have?

- A. None
- B. Exactly one
- C. Exactly two
- D. Infinitely many

309. Systems of linear equations · Medium · MC

How many solutions does the system $y = 4x + 2$ and $y = 4x + 6$ have?

- A. None
- B. Exactly one
- C. Exactly two
- D. Infinitely many

310. Systems of linear equations · Medium · MC

How many solutions does the system $y = 3x + 1$ and $3y = 9x + 3$ have?

- A. Exactly two
- B. Infinitely many
- C. None
- D. Exactly one

311. Solve for a variable · Easy · MC

The perimeter of a rectangle is 46 and the width is 10. What is the length?

- A. 23
- B. 10
- C. 13
- D. 36

312. Solve for a variable · Easy · SPR

The perimeter of a rectangle is 24 and the width is 7. What is the length?

313. Solve for a variable · Easy · SPR

The perimeter of a rectangle is 42 and the width is 6. What is the length?

314. Solve for a variable · Easy · SPR

The perimeter of a rectangle is 16 and the width is 3. What is the length?

315. Solve for a variable · Easy · MC

The perimeter of a rectangle is 44 and the width is 10. What is the length?

A. 22

B. 12

C. 34

D. 10

316. Solve for a variable · Easy · MC

The perimeter of a rectangle is 34 and the width is 10. What is the length?

A. 7

B. 10

C. 24

D. 17

317. Solve for a variable · Easy · SPR

The perimeter of a rectangle is 30 and the width is 5. What is the length?

318. Solve for a variable · Easy · MC

The perimeter of a rectangle is 22 and the width is 7. What is the length?

A. 11

B. 15

C. 4

D. 7

319. Solve for a variable · Easy · MC

The perimeter of a rectangle is 44 and the width is 7. What is the length?

A. 22

B. 37

C. 15

D. 7

320. Solve for a variable · Easy · MC

The perimeter of a rectangle is 34 and the width is 3. What is the length?

A. 14

B. 17

C. 3

D. 31

321. Solve for a variable · Easy · MC

The perimeter of a rectangle is 28 and the width is 4. What is the length?

- A. 14
- B. 4
- C. 24
- D. 10

322. Solve for a variable · Easy · MC

The perimeter of a rectangle is 42 and the width is 9. What is the length?

- A. 21
- B. 33
- C. 9
- D. 12

323. Solve for a variable · Easy · MC

The perimeter of a rectangle is 14 and the width is 2. What is the length?

- A. 7
- B. 12
- C. 2
- D. 5

324. Solve for a variable · Easy · MC

The perimeter of a rectangle is 28 and the width is 6. What is the length?

- A. 6
- B. 22
- C. 14
- D. 8

325. Solve for a variable · Easy · SPR

The perimeter of a rectangle is 18 and the width is 5. What is the length?

326. Solve for a variable · Medium · SPR

The perimeter of a rectangle is 22 and the width is 4. What is the length?

327. Solve for a variable · Medium · MC

The perimeter of a rectangle is 24 and the width is 6. What is the length?

- A. 9
- B. 12
- C. 6
- D. 18

328. Solve for a variable · Medium · MC

The perimeter of a rectangle is 32 and the width is 6. What is the length?

- A. 16
- B. 10
- C. 26
- D. 6

329. Solve for a variable · Medium · MC

The perimeter of a rectangle is 26 and the width is 6. What is the length?

- A. 20
- B. 7
- C. 6
- D. 13

330. Solve for a variable · Medium · MC

The perimeter of a rectangle is 26 and the width is 4. What is the length?

- A. 13
- B. 4
- C. 22
- D. 9

331. Solve for a variable · Medium · SPR

The perimeter of a rectangle is 26 and the width is 9. What is the length?

332. Solve for a variable · Medium · MC

The perimeter of a rectangle is 28 and the width is 2. What is the length?

- A. 2
- B. 26
- C. 14
- D. 12

333. Solve for a variable · Medium · SPR

The perimeter of a rectangle is 36 and the width is 4. What is the length?

334. Solve for a variable · Medium · SPR

The perimeter of a rectangle is 30 and the width is 4. What is the length?

335. Solve for a variable · Medium · MC

The perimeter of a rectangle is 26 and the width is 5. What is the length?

- A. 5
- B. 13
- C. 8
- D. 21

336. Solve for a variable · Medium · MC

The perimeter of a rectangle is 22 and the width is 6. What is the length?

- A. 16
- B. 5
- C. 11
- D. 6

337. Solve for a variable · Medium · MC

The perimeter of a rectangle is 38 and the width is 7. What is the length?

- A. 31
- B. 12
- C. 19
- D. 7

338. Solve for a variable · Medium · MC

The perimeter of a rectangle is 30 and the width is 2. What is the length?

- A. 13
- B. 2
- C. 28
- D. 15

339. Solve for a variable · Medium · MC

The perimeter of a rectangle is 22 and the width is 5. What is the length?

- A. 6
- B. 5
- C. 11
- D. 17

340. Solve for a variable · Medium · SPR

The perimeter of a rectangle is 44 and the width is 9. What is the length?

341. Solve for a variable · Hard · MC

A car travels 36 miles in 2 hours at a constant speed. What is the speed in miles per hour?

- A. 72
- B. 38
- C. 2
- D. 18

342. Solve for a variable · Hard · MC

A car travels 48 miles in 6 hours at a constant speed. What is the speed in miles per hour?

- A. 288
- B. 8
- C. 6
- D. 54

343. Solve for a variable · Hard · MC

A car travels 28 miles in 2 hours at a constant speed. What is the speed in miles per hour?

- A. 14
- B. 56
- C. 2
- D. 30

344. Solve for a variable · Hard · MC

A car travels 144 miles in 6 hours at a constant speed. What is the speed in miles per hour?

- A. 150
- B. 24
- C. 864
- D. 6

345. Solve for a variable · Hard · SPR

A car travels 33 miles in 3 hours at a constant speed. What is the speed in miles per hour?

346. Solve for a variable · Hard · MC

A car travels 50 miles in 5 hours at a constant speed. What is the speed in miles per hour?

- A. 250
- B. 55
- C. 10
- D. 5

347. Solve for a variable · Hard · SPR

A car travels 36 miles in 3 hours at a constant speed. What is the speed in miles per hour?

348. Solve for a variable · Hard · MC

A car travels 65 miles in 5 hours at a constant speed. What is the speed in miles per hour?

- A. 70
- B. 325
- C. 13
- D. 5

349. Solve for a variable · Hard · MC

A car travels 95 miles in 5 hours at a constant speed. What is the speed in miles per hour?

- A. 475
- B. 100
- C. 19
- D. 5

350. Solve for a variable · Hard · SPR

A car travels 50 miles in 2 hours at a constant speed. What is the speed in miles per hour?

Advanced Math

351. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 6)(x + 5)$?

- A. $x^2 + 11x + 30$
- B. $x^2 + 30x + 11$
- C. $x^2 + 11x + 12$
- D. $x^2 - 11x + 30$

352. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 1)(x + 7)$?

- A. $x^2 - 8x + 7$
- B. $x^2 + 8x + 7$
- C. $x^2 + 7x + 8$
- D. $x^2 + 8x + 9$

353. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 8)(x + 6)$?

- A. $x^2 + 14x + 48$
- B. $x^2 + 48x + 14$
- C. $x^2 + 14x + 15$
- D. $x^2 - 14x + 48$

354. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 8)(x + 5)$?

- A. $x^2 + 13x + 14$
- B. $x^2 - 13x + 40$
- C. $x^2 + 13x + 40$
- D. $x^2 + 40x + 13$

355. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 6)(x + 2)$?

- A. $x^2 + 8x + 12$
- B. $x^2 + 12x + 8$
- C. $x^2 + 8x + 9$
- D. $x^2 - 8x + 12$

356. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 3)(x + 4)$?

- A. $x^2 + 7x + 12$
- B. $x^2 + 12x + 7$
- C. $x^2 + 7x + 8$
- D. $x^2 - 7x + 12$

357. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 4)(x + 7)$?

- A. $x^2 + 11x + 12$
- B. $x^2 - 11x + 28$
- C. $x^2 + 11x + 28$
- D. $x^2 + 28x + 11$

358. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 6)(x + 1)$?

- A. $x^2 + 6x + 7$
- B. $x^2 + 7x + 8$
- C. $x^2 - 7x + 6$
- D. $x^2 + 7x + 6$

359. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 4)(x + 3)$?

- A. $x^2 + 7x + 12$
- B. $x^2 + 12x + 7$
- C. $x^2 + 7x + 8$
- D. $x^2 - 7x + 12$

360. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 2)(x + 5)$?

- A. $x^2 + 7x + 10$
- B. $x^2 + 10x + 7$
- C. $x^2 + 7x + 8$
- D. $x^2 - 7x + 10$

361. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 1)(x + 5)$?

- A. $x^2 + 6x + 5$
- B. $x^2 + 5x + 6$
- C. $x^2 + 6x + 7$
- D. $x^2 - 6x + 5$

362. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 7)(x + 2)$?

- A. $x^2 + 9x + 10$
- B. $x^2 - 9x + 14$
- C. $x^2 + 9x + 14$
- D. $x^2 + 14x + 9$

363. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 1)(x + 6)$?

- A. $x^2 + 7x + 8$
- B. $x^2 - 7x + 6$
- C. $x^2 + 7x + 6$
- D. $x^2 + 6x + 7$

364. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 1)(x + 3)$?

- A. $x^2 - 4x + 3$
- B. $x^2 + 4x + 3$
- C. $x^2 + 3x + 4$
- D. $x^2 + 4x + 5$

365. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 7)(x + 5)$?

- A. $x^2 + 12x + 13$
- B. $x^2 - 12x + 35$
- C. $x^2 + 12x + 35$
- D. $x^2 + 35x + 12$

366. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 8)(x + 4)$?

- A. $x^2 + 32x + 12$
- B. $x^2 + 12x + 13$
- C. $x^2 - 12x + 32$
- D. $x^2 + 12x + 32$

367. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 5)(x + 2)$?

- A. $x^2 + 7x + 8$
- B. $x^2 - 7x + 10$
- C. $x^2 + 7x + 10$
- D. $x^2 + 10x + 7$

368. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 4)(x + 2)$?

- A. $x^2 - 6x + 8$
- B. $x^2 + 6x + 8$
- C. $x^2 + 8x + 6$
- D. $x^2 + 6x + 7$

369. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 5)(x + 1)$?

- A. $x^2 - 6x + 5$
- B. $x^2 + 6x + 5$
- C. $x^2 + 5x + 6$
- D. $x^2 + 6x + 7$

370. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 4)(x + 4)$?

- A. $x^2 - 8x + 16$
- B. $x^2 + 8x + 16$
- C. $x^2 + 16x + 8$
- D. $x^2 + 8x + 9$

371. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 8)(x + 7)$?

- A. $x^2 + 15x + 16$
- B. $x^2 - 15x + 56$
- C. $x^2 + 15x + 56$
- D. $x^2 + 56x + 15$

372. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 4)(x + 8)$?

- A. $x^2 - 12x + 32$
- B. $x^2 + 12x + 32$
- C. $x^2 + 32x + 12$
- D. $x^2 + 12x + 13$

373. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 2)(x + 8)$?

- A. $x^2 - 10x + 16$
- B. $x^2 + 10x + 16$
- C. $x^2 + 16x + 10$
- D. $x^2 + 10x + 11$

374. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 2)(x + 7)$?

- A. $x^2 + 9x + 14$
- B. $x^2 + 14x + 9$
- C. $x^2 + 9x + 10$
- D. $x^2 - 9x + 14$

375. Equivalent expressions · Easy · MC

Which expression is equivalent to $(x + 6)(x + 7)$?

- A. $x^2 + 13x + 42$
- B. $x^2 + 42x + 13$
- C. $x^2 + 13x + 14$
- D. $x^2 - 13x + 42$

376. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 6x + 5$?

- A. $(x - 1)(x - 5)$
- B. $(x + 1)(x + 5)$
- C. $(x + 6)(x + 5)$
- D. $(x + 1)(x - 5)$

377. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 9x + 18$?

- A. $(x + 3)(x - 6)$
- B. $(x - 3)(x - 6)$
- C. $(x + 3)(x + 6)$
- D. $(x + 9)(x + 18)$

378. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 7x + 6$?

- A. $(x + 1)(x + 6)$
- B. $(x + 7)(x + 6)$
- C. $(x + 1)(x - 6)$
- D. $(x - 1)(x - 6)$

379. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 9x + 20$?

- A. $(x + 5)(x + 4)$
- B. $(x + 9)(x + 20)$
- C. $(x + 5)(x - 4)$
- D. $(x - 5)(x - 4)$

380. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 10x + 16$?

- A. $(x - 2)(x - 8)$
- B. $(x + 2)(x + 8)$
- C. $(x + 10)(x + 16)$
- D. $(x + 2)(x - 8)$

381. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 14x + 49$?

- A. $(x + 14)(x + 49)$
- B. $(x + 7)(x - 7)$
- C. $(x - 7)(x - 7)$
- D. $(x + 7)(x + 7)$

382. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 4x + 4$?

- A. $(x + 2)(x - 2)$
- B. $(x - 2)(x - 2)$
- C. $(x + 2)(x + 2)$
- D. $(x + 4)(x + 4)$

383. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 10x + 21$?

- A. $(x + 10)(x + 21)$
- B. $(x + 7)(x - 3)$
- C. $(x - 7)(x - 3)$
- D. $(x + 7)(x + 3)$

384. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 3x + 2$?

- A. $(x - 1)(x - 2)$
- B. $(x + 1)(x + 2)$
- C. $(x + 3)(x + 2)$
- D. $(x + 1)(x - 2)$

385. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 16x + 64$?

- A. $(x - 8)(x - 8)$
- B. $(x + 8)(x + 8)$
- C. $(x + 16)(x + 64)$
- D. $(x + 8)(x - 8)$

386. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 6x + 8$?

- A. $(x + 6)(x + 8)$
- B. $(x + 2)(x - 4)$
- C. $(x - 2)(x - 4)$
- D. $(x + 2)(x + 4)$

387. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 8x + 7$?

- A. $(x + 7)(x - 1)$
- B. $(x - 7)(x - 1)$
- C. $(x + 7)(x + 1)$
- D. $(x + 8)(x + 7)$

388. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 15x + 56$?

- A. $(x - 8)(x - 7)$
- B. $(x + 8)(x + 7)$
- C. $(x + 15)(x + 56)$
- D. $(x + 8)(x - 7)$

389. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 7x + 10$?

- A. $(x + 7)(x + 10)$
- B. $(x + 2)(x - 5)$
- C. $(x - 2)(x - 5)$
- D. $(x + 2)(x + 5)$

390. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 4x + 3$?

- A. $(x + 3)(x + 1)$
- B. $(x + 4)(x + 3)$
- C. $(x + 3)(x - 1)$
- D. $(x - 3)(x - 1)$

391. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 12x + 32$?

- A. $(x + 12)(x + 32)$
- B. $(x + 4)(x - 8)$
- C. $(x - 4)(x - 8)$
- D. $(x + 4)(x + 8)$

392. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 11x + 24$?

- A. $(x + 3)(x + 8)$
- B. $(x + 11)(x + 24)$
- C. $(x + 3)(x - 8)$
- D. $(x - 3)(x - 8)$

393. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 5x + 6$?

- A. $(x + 3)(x - 2)$
- B. $(x - 3)(x - 2)$
- C. $(x + 3)(x + 2)$
- D. $(x + 5)(x + 6)$

394. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 12x + 35$?

- A. $(x - 5)(x - 7)$
- B. $(x + 5)(x + 7)$
- C. $(x + 12)(x + 35)$
- D. $(x + 5)(x - 7)$

395. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 7x + 12$?

- A. $(x - 3)(x - 4)$
- B. $(x + 3)(x + 4)$
- C. $(x + 7)(x + 12)$
- D. $(x + 3)(x - 4)$

396. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 13x + 42$?

- A. $(x + 7)(x + 6)$
- B. $(x + 13)(x + 42)$
- C. $(x + 7)(x - 6)$
- D. $(x - 7)(x - 6)$

397. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 9x + 14$?

- A. $(x - 2)(x - 7)$
- B. $(x + 2)(x + 7)$
- C. $(x + 9)(x + 14)$
- D. $(x + 2)(x - 7)$

398. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 8x + 12$?

- A. $(x + 8)(x + 12)$
- B. $(x + 2)(x - 6)$
- C. $(x - 2)(x - 6)$
- D. $(x + 2)(x + 6)$

399. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 10x + 24$?

- A. $(x - 4)(x - 6)$
- B. $(x + 4)(x + 6)$
- C. $(x + 10)(x + 24)$
- D. $(x + 4)(x - 6)$

400. Equivalent expressions · Medium · MC

Which expression is equivalent to $x^2 + 9x + 8$?

- A. $(x + 1)(x - 8)$
- B. $(x - 1)(x - 8)$
- C. $(x + 1)(x + 8)$
- D. $(x + 9)(x + 8)$

401. Nonlinear equations · Easy · SPR

What is the sum of the solutions of $x^2 + 3x - 10 = 0$?

402. Nonlinear equations · Easy · SPR

What is the sum of the solutions of $x^2 - 6x - 7 = 0$?

403. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 + 1x = 0$?

- A. 0
- B. -2
- C. 1
- D. -1

404. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 - 10x + 24 = 0$?

- A. 4
- B. 24
- C. 10
- D. 6

405. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 - 5x + 4 = 0$?

- A. 4
- B. 5
- C. 1
- D. 3

406. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 - 4 = 0$?

- A. -4
- B. 0
- C. 2
- D. -2

407. Nonlinear equations · Easy · SPR

What is the sum of the solutions of $x^2 - 3x - 28 = 0$?

408. Nonlinear equations · Easy · SPR

What is the sum of the solutions of $x^2 - 11x + 28 = 0$?

409. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 - 6x + 5 = 0$?

- A. 6
- B. 4
- C. 1
- D. 5

410. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 + 7x + 10 = 0$?

- A. -5
- B. -2
- C. -7
- D. 10

411. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 + 5x + 4 = 0$?

- A. -4
- B. -5
- C. 4
- D. -1

412. Nonlinear equations · Easy · SPR

What is the sum of the solutions of $x^2 - 7x + 10 = 0$?

413. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 - 10x + 9 = 0$?

- A. 9
- B. 8
- C. 10
- D. 1

414. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 - 6x = 0$?

- A. 6
- B. 7
- C. 5
- D. 0

415. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 + 1x - 6 = 0$?

- A. 2
- B. -3
- C. -1
- D. -6

416. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 - 4x - 45 = 0$?

- A. -5
- B. 9
- C. -45
- D. 4

417. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 - 16 = 0$?

- A. 4
- B. -4
- C. 0
- D. -16

418. Nonlinear equations · Easy · SPR

What is the sum of the solutions of $x^2 - 1x - 12 = 0$?

419. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 - 9x + 20 = 0$?

- A. 4
- B. 20
- C. 9
- D. 5

420. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 + 2x - 8 = 0$?

- A. -2
- B. 2
- C. -8
- D. -4

421. Nonlinear equations · Easy · SPR

What is the sum of the solutions of $x^2 + 4x = 0$?

422. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 + 6x + 5 = 0$?

- A. -5
- B. 5
- C. -6
- D. -1

423. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 - 5x - 6 = 0$?

- A. -6
- B. -1
- C. 6
- D. 5

424. Nonlinear equations · Easy · SPR

What is the sum of the solutions of $x^2 - 9x = 0$?

425. Nonlinear equations · Easy · MC

What is the sum of the solutions of $x^2 - 3x - 4 = 0$?

- A. 3
- B. 4
- C. -1
- D. -4

426. Nonlinear equations · Medium · SPR

What is the greater solution of $x^2 - 15x + 54 = 0$?

427. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 11x + 24 = 0$?

- A. 11
- B. 24
- C. 3
- D. 8

428. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 7x + 10 = 0$?

- A. 2
- B. 7
- C. 5
- D. 10

429. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 + 1x - 20 = 0$?

- A. -5
- B. -1
- C. 4
- D. -20

430. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 2x - 15 = 0$?

- A. 5
- B. 2
- C. -3
- D. -15

431. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 + 5x + 6 = 0$?

- A. -2
- B. -3
- C. 6
- D. -5

432. Nonlinear equations · Medium · SPR

What is the greater solution of $x^2 - 7x + 12 = 0$?

433. Nonlinear equations · Medium · SPR

What is the greater solution of $x^2 + 7x + 10 = 0$?

434. Nonlinear equations · Medium · SPR

What is the greater solution of $x^2 - 1x - 6 = 0$?

435. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 2x - 8 = 0$?

- A. -8
- B. 2
- C. 4
- D. -2

436. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 6x = 0$?

- A. 6
- B. 5
- C. 7
- D. 0

437. Nonlinear equations · Medium · SPR

What is the greater solution of $x^2 - 15x + 56 = 0$?

438. Nonlinear equations · Medium · SPR

What is the greater solution of $x^2 - 9x + 14 = 0$?

439. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 10x + 21 = 0$?

- A. 3
- B. 10
- C. 21
- D. 7

440. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 5x - 6 = 0$?

- A. -1
- B. 5
- C. -6
- D. 6

441. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 + 4x - 5 = 0$?

- A. 2
- B. 1
- C. -5
- D. -4

442. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 + 2x - 3 = 0$?

- A. -3
- B. 1
- C. -2
- D. 2

443. Nonlinear equations · Medium · SPR

What is the greater solution of $x^2 - 13x + 40 = 0$?

444. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 2x = 0$?

- A. 1
- B. 0
- C. 2
- D. 3

445. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 5x - 14 = 0$?

- A. -14
- B. 5
- C. -2
- D. 7

446. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 7x - 8 = 0$?

- A. 7
- B. -8
- C. -1
- D. 8

447. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 3x + 2 = 0$?

- A. 2
- B. 3
- C. 1
- D. 4

448. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 1x - 30 = 0$?

- A. 6
- B. -5
- C. -30
- D. 1

449. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 2x - 35 = 0$?

- A. 7
- B. -5
- C. 2
- D. -35

450. Nonlinear equations · Medium · MC

What is the greater solution of $x^2 - 6x + 8 = 0$?

- A. 8
- B. 6
- C. 4
- D. 2

451. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 10x + 32$ is defined for all real x . What is the minimum value of f ?

- A. -5
- B. 7
- C. 5
- D. 32

452. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 2x + -3$ is defined for all real x . What is the minimum value of f ?

- A. -1
- B. 1
- C. -4
- D. -3

453. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 8x + 10$ is defined for all real x . What is the minimum value of f ?

- A. 4
- B. 10
- C. -4
- D. -6

454. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 14x + 49$ is defined for all real x . What is the minimum value of f ?

- A. -7
- B. 0
- C. 7
- D. 49

455. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 4x + 10$ is defined for all real x . What is the minimum value of f ?

- A. 6
- B. 2
- C. 10
- D. -2

456. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 4x + 12$ is defined for all real x . What is the minimum value of f ?

- A. 8
- B. -2
- C. 12
- D. 2

457. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 16x + 59$ is defined for all real x . What is the minimum value of f ?

- A. 8
- B. 59
- C. -8
- D. -5

458. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 2x - 2$ is defined for all real x . What is the minimum value of f ?

- A. -1
- B. -2
- C. 1
- D. -3

459. Nonlinear functions · Hard · SPR

The function $f(x) = x^2 - 2x + 8$ is defined for all real x . What is the minimum value of f ?

460. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 14x + 46$ is defined for all real x . What is the minimum value of f ?

- A. 46
- B. -3
- C. -7
- D. 7

461. Nonlinear functions · Hard · SPR

The function $f(x) = x^2 - 4x + 13$ is defined for all real x . What is the minimum value of f ?

462. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 8x + 22$ is defined for all real x . What is the minimum value of f ?

- A. -4
- B. 4
- C. 22
- D. 6

463. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 12x + 32$ is defined for all real x . What is the minimum value of f ?

- A. 32
- B. -4
- C. 6
- D. -6

464. Nonlinear functions · Hard · SPR

The function $f(x) = x^2 - 16x + 61$ is defined for all real x . What is the minimum value of f ?

465. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 16x + 66$ is defined for all real x . What is the minimum value of f ?

- A. 2
- B. 66
- C. 8
- D. -8

466. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 8x + 23$ is defined for all real x . What is the minimum value of f ?

- A. 7
- B. 23
- C. -4
- D. 4

467. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 8x + 20$ is defined for all real x . What is the minimum value of f ?

- A. 8
- B. 20
- C. 4
- D. -4

468. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 14x + 56$ is defined for all real x . What is the minimum value of f ?

- A. 7
- B. -7
- C. 14
- D. 56

469. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 6x + 4$ is defined for all real x . What is the minimum value of f ?

- A. 4
- B. -3
- C. -5
- D. 3

470. Nonlinear functions · Hard · MC

The function $f(x) = x^2 - 6x + 8$ is defined for all real x . What is the minimum value of f ?

- A. -3
- B. 8
- C. -1
- D. 3

471. Nonlinear functions · Easy · MC

If $f(t) = 2 \cdot 3^t$, what is the value of $f(2)$?

- A. 18
- B. 11
- C. 12
- D. 9

472. Nonlinear functions · Easy · SPR

If $f(t) = 8 \cdot 4^t$, what is the value of $f(4)$?

473. Nonlinear functions · Easy · SPR

If $f(t) = 8 \cdot 2^t$, what is the value of $f(3)$?

474. Nonlinear functions · Easy · MC

If $f(t) = 3 \cdot 3^t$, what is the value of $f(2)$?

- A. 9
- B. 12
- C. 27
- D. 18

475. Nonlinear functions · Easy · MC

If $f(t) = 10 \cdot 3^t$, what is the value of $f(3)$?

- A. 270
- B. 27
- C. 37
- D. 90

476. Nonlinear functions · Easy · SPR

If $f(t) = 11 \cdot 2^t$, what is the value of $f(2)$?

477. Nonlinear functions · Easy · SPR

If $f(t) = 4 \cdot 3^t$, what is the value of $f(4)$?

478. Nonlinear functions · Easy · MC

If $f(t) = 9 \cdot 2^t$, what is the value of $f(3)$?

- A. 8
- B. 54
- C. 72
- D. 17

479. Nonlinear functions · Easy · SPR

If $f(t) = 10 \cdot 4^t$, what is the value of $f(3)$?

480. Nonlinear functions · Easy · SPR

If $f(t) = 7 \cdot 3^t$, what is the value of $f(2)$?

481. Nonlinear functions · Easy · MC

If $f(t) = 8 \cdot 2^t$, what is the value of $f(2)$?

- A. 12
- B. 32
- C. 4
- D. 18

482. Nonlinear functions · Easy · MC

If $f(t) = 12 \cdot 4^t$, what is the value of $f(4)$?

- A. 192
- B. 256
- C. 3072
- D. 268

483. Nonlinear functions · Easy · MC

If $f(t) = 5 \cdot 3^t$, what is the value of $f(4)$?

- A. 81
- B. 86
- C. 60
- D. 405

484. Nonlinear functions · Easy · MC

If $f(t) = 8 \cdot 2^t$, what is the value of $f(4)$?

- A. 64
- B. 128
- C. 16
- D. 24

485. Nonlinear functions · Easy · SPR

If $f(t) = 2 \cdot 2^t$, what is the value of $f(3)$?

486. Nonlinear functions · Medium · MC

A population is modeled by $P(t) = 200 \cdot 3^t$, where t is the number of years after the count begins. What is $P(2)$?

- A. 1200
- B. 9
- C. 1800
- D. 206

487. Nonlinear functions · Medium · MC

A population is modeled by $P(t) = 20 \cdot 3^t$, where t is the number of years after the count begins. What is $P(3)$?

- A. 29
- B. 27
- C. 180
- D. 540

488. Nonlinear functions · Medium · MC

A population is modeled by $P(t) = 40 \cdot 3^t$, where t is the number of years after the count begins. What is $P(2)$?

- A. 46
- B. 240
- C. 9
- D. 360

489. Nonlinear functions · Medium · MC

A population is modeled by $P(t) = 50 \cdot 2^t$, where t is the number of years after the count begins. What is $P(4)$?

- A. 800
- B. 16
- C. 58
- D. 400

490. Nonlinear functions · Medium · SPR

A population is modeled by $P(t) = 125 \cdot 2^t$, where t is the number of years after the count begins. What is $P(3)$?

491. Nonlinear functions · Medium · SPR

A population is modeled by $P(t) = 100 \cdot 2^t$, where t is the number of years after the count begins. What is $P(3)$?

492. Nonlinear functions · Medium · MC

A population is modeled by $P(t) = 250 \cdot 3^t$, where t is the number of years after the count begins. What is $P(2)$?

- A. 9
- B. 1500
- C. 256
- D. 2250

493. Nonlinear functions · Medium · SPR

A population is modeled by $P(t) = 80 \cdot 2^t$, where t is the number of years after the count begins. What is $P(3)$?

494. Nonlinear functions · Medium · MC

A population is modeled by $P(t) = 250 \cdot 3^t$, where t is the number of years after the count begins. What is $P(4)$?

- A. 262
- B. 3000
- C. 81
- D. 20250

495. Nonlinear functions · Medium · SPR

A population is modeled by $P(t) = 200 \cdot 2^t$, where t is the number of years after the count begins. What is $P(3)$?

496. Nonlinear functions · Medium · MC

A population is modeled by $P(t) = 80 \cdot 3^t$, where t is the number of years after the count begins. What is $P(2)$?

- A. 9
- B. 86
- C. 480
- D. 720

497. Nonlinear functions · Medium · MC

A population is modeled by $P(t) = 125 \cdot 3^t$, where t is the number of years after the count begins. What is $P(4)$?

- A. 10125
- B. 1500
- C. 81
- D. 137

498. Nonlinear functions · Medium · MC

A population is modeled by $P(t) = 20 \cdot 3^t$, where t is the number of years after the count begins. What is $P(2)$?

- A. 120
- B. 180
- C. 26
- D. 9

499. Nonlinear functions · Medium · MC

A population is modeled by $P(t) = 250 \cdot 2^t$, where t is the number of years after the count begins. What is $P(3)$?

- A. 8
- B. 1500
- C. 256
- D. 2000

500. Nonlinear functions · Medium · MC

A population is modeled by $P(t) = 125 \cdot 2^t$, where t is the number of years after the count begins. What is $P(4)$?

- A. 133
- B. 16
- C. 2000
- D. 1000

501. Nonlinear functions · Hard · SPR

A quantity grows by the model $P(t) = a \cdot 3^t$. If $P(2) = 162$, what is the value of a ?

502. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 4^t$. If $P(2) = 112$, what is the value of a ?

- A. 56
- B. 16
- C. 7
- D. 28

503. Nonlinear functions · Hard · SPR

A quantity grows by the model $P(t) = a \cdot 4^t$. If $P(2) = 192$, what is the value of a ?

504. Nonlinear functions · Hard · SPR

A quantity grows by the model $P(t) = a \cdot 3^t$. If $P(2) = 27$, what is the value of a ?

505. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 2^t$. If $P(2) = 72$, what is the value of a ?

- A. 4
- B. 72
- C. 18
- D. 36

506. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 3^t$. If $P(2) = 171$, what is the value of a ?

- A. 9
- B. 114
- C. 57
- D. 19

507. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 2^t$. If $P(2) = 32$, what is the value of a ?

- A. 16
- B. 4
- C. 32
- D. 8

508. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 2^t$. If $P(2) = 28$, what is the value of a ?

- A. 14
- B. 4
- C. 28
- D. 7

509. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 4^t$. If $P(2) = 288$, what is the value of a ?

- A. 72
- B. 16
- C. 144
- D. 18

510. Nonlinear functions · Hard · MC

A quantity grows by the model $P(t) = a \cdot 3^t$. If $P(2) = 90$, what is the value of a ?

- A. 30
- B. 9
- C. 10
- D. 60

511. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^2 \cdot x^5$ for $x \neq 0$?

- A. x^3
- B. $10x$
- C. x^7
- D. x^{10}

512. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^2 \cdot x^4$ for $x \neq 0$?

- A. x^2
- B. $8x$
- C. x^6
- D. x^8

513. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^4 \cdot x^3$ for $x \neq 0$?

- A. x^7
- B. x^{12}
- C. x^1
- D. $12x$

514. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^2 \cdot x^3$ for $x \neq 0$?

- A. x^5
- B. x^6
- C. x^1
- D. $6x$

515. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^3 \cdot x^5$ for $x \neq 0$?

- A. $15x$
- B. x^8
- C. x^{15}
- D. x^2

516. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^4 \cdot x^5$ for $x \neq 0$?

- A. x^{20}
- B. x^1
- C. $20x$
- D. x^9

517. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^4 \cdot x^2$ for $x \neq 0$?

- A. x^2
- B. $8x$
- C. x^6
- D. x^8

518. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^5 \cdot x^2$ for $x \neq 0$?

- A. $10x$
- B. x^7
- C. x^{10}
- D. x^3

519. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^5 \cdot x^3$ for $x \neq 0$?

- A. $15x$
- B. x^8
- C. x^{15}
- D. x^2

520. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^6 \cdot x^5$ for $x \neq 0$?

- A. x^{30}
- B. x^1
- C. $30x$
- D. x^{11}

521. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^5 \cdot x^5$ for $x \neq 0$?

- A. x^{25}
- B. x^0
- C. $25x$
- D. x^{10}

522. Equivalent expressions · Easy · MC

Which expression is equivalent to $x^4 \cdot x^4$ for $x \neq 0$?

- A. x^8
- B. x^{16}
- C. x^0
- D. $16x$

523. Equivalent expressions · Medium · MC

Which expression is equivalent to $(x^4)^5$?

- A. x^4
- B. $20x$
- C. x^{20}
- D. x^9

524. Equivalent expressions · Medium · MC

Which expression is equivalent to $(x^5)^4$?

- A. $20x$
- B. x^{20}
- C. x^9
- D. x^5

525. Equivalent expressions · Medium · MC

Which expression is equivalent to $(x^4)^4$?

- A. x^8
- B. x^4
- C. $16x$
- D. x^{16}

526. Equivalent expressions · Medium · MC

Which expression is equivalent to $(x^6)^3$?

- A. x^9
- B. x^6
- C. $18x$
- D. x^{18}

527. Equivalent expressions · Medium · MC

Which expression is equivalent to $(x^3)^4$?

- A. $12x$
- B. x^{12}
- C. x^7
- D. x^3

528. Equivalent expressions · Medium · MC

Which expression is equivalent to $(x^3)^5$?

- A. x^3
- B. $15x$
- C. x^{15}
- D. x^8

529. Equivalent expressions · Medium · MC

Which expression is equivalent to $(x^2)^3$?

- A. x^5
- B. x^2
- C. $6x$
- D. x^6

530. Equivalent expressions · Medium · MC

Which expression is equivalent to $(x^3)^2$?

- A. x^6
- B. x^5
- C. x^3
- D. $6x$

531. Equivalent expressions · Medium · MC

Which expression is equivalent to $(x^5)^5$?

- A. x^{25}
- B. x^{10}
- C. x^5
- D. $25x$

532. Equivalent expressions · Medium · MC

Which expression is equivalent to $(x^4)^3$?

- A. x^{12}
- B. x^7
- C. x^4
- D. $12x$

533. Equivalent expressions · Medium · MC

Which expression is equivalent to $(x^6)^4$?

- A. x^{10}
- B. x^6
- C. $24x$
- D. x^{24}

534. Equivalent expressions · Medium · MC

Which expression is equivalent to $(x^2)^4$?

- A. x^2
- B. $8x$
- C. x^8
- D. x^6

535. Equivalent expressions · Medium · MC

Which expression is equivalent to $(x^2)^5$?

- A. $10x$
- B. x^{10}
- C. x^7
- D. x^2

536. Nonlinear functions · Easy · SPR

If $f(x) = 4x + 3$, what is the value of $f(-3)$?

537. Nonlinear functions · Easy · SPR

If $f(x) = 6x - 5$, what is the value of $f(4)$?

538. Nonlinear functions · Easy · MC

If $f(x) = 6x + 3$, what is the value of $f(2)$?

- A. 15
- B. 9
- C. 11
- D. 30

539. Nonlinear functions · Easy · SPR

If $f(x) = 6x - 4$, what is the value of $f(1)$?

540. Nonlinear functions · Easy · SPR

If $f(x) = 3x + 6$, what is the value of $f(-1)$?

541. Nonlinear functions · Easy · MC

If $f(x) = 2x - 4$, what is the value of $f(-3)$?

- A. -10
- B. -5
- C. -14
- D. -2

542. Nonlinear functions · Easy · MC

If $f(x) = 5x + 6$, what is the value of $f(3)$?

- A. 45
- B. 14
- C. 21
- D. 9

543. Nonlinear functions · Easy · SPR

If $f(x) = 4x + 5$, what is the value of $f(3)$?

544. Nonlinear functions · Easy · SPR

If $f(x) = 6x$, what is the value of $f(0)$?

545. Nonlinear functions · Easy · MC

If $f(x) = 3x + 1$, what is the value of $f(-1)$?

- A. 0
- B. -4
- C. 3
- D. -2

546. Nonlinear functions · Easy · SPR

If $f(x) = 2x + 4$, what is the value of $f(6)$?

547. Nonlinear functions · Easy · MC

If $f(x) = 6x - 4$, what is the value of $f(-3)$?

- A. -22
- B. -42
- C. -14
- D. -1

548. Nonlinear functions · Easy · SPR

If $f(x) = 3x + 7$, what is the value of $f(-1)$?

549. Nonlinear functions · Easy · MC

If $f(x) = 4x + 5$, what is the value of $f(0)$?

- A. 20
- B. 9
- C. 5
- D. -5

550. Nonlinear functions · Easy · MC

If $f(x) = 2x - 3$, what is the value of $f(2)$?

- A. 7
- B. 1
- C. -2
- D. 2

551. Nonlinear functions · Easy · MC

If $f(x) = 2x - 5$, what is the value of $f(-3)$?

- A. -16
- B. -11
- C. -6
- D. -1

552. Nonlinear functions · Easy · MC

If $f(x) = 4x - 2$, what is the value of $f(-3)$?

- A. -1
- B. -20
- C. -14
- D. -10

553. Nonlinear functions · Easy · MC

If $f(x) = 5x - 3$, what is the value of $f(6)$?

- A. 15
- B. 8
- C. 33
- D. 27

554. Nonlinear functions · Easy · SPR

If $f(x) = 4x + 7$, what is the value of $f(6)$?

555. Nonlinear functions · Easy · SPR

If $f(x) = 6x$, what is the value of $f(7)$?

556. Nonlinear functions · Medium · MC

If $f(x) = 4x + 5$ and $g(x) = 2x + 3$, what is the value of $f(g(-1))$?

- A. -8
- B. 9
- C. 2
- D. 10

557. Nonlinear functions · Medium · MC

If $f(x) = 2x + 5$ and $g(x) = 2x$, what is the value of $f(g(2))$?

- A. 8
- B. 12
- C. 13
- D. 14

558. Nonlinear functions · Medium · MC

If $f(x) = 3x + 3$ and $g(x) = 4x + 1$, what is the value of $f(g(2))$?

- A. 24
- B. 18
- C. 31
- D. 30

559. Nonlinear functions · Medium · MC

If $f(x) = 3x + 1$ and $g(x) = 3x - 1$, what is the value of $f(g(-2))$?

- A. -20
- B. -19
- C. -18
- D. -12

560. Nonlinear functions · Medium · MC

If $f(x) = 3x - 4$ and $g(x) = 4x - 1$, what is the value of $f(g(-2))$?

- A. -31
- B. -30
- C. -19
- D. -24

561. Nonlinear functions · Medium · MC

If $f(x) = 3x - 4$ and $g(x) = 4x + 4$, what is the value of $f(g(3))$?

- A. 36
- B. 21
- C. 44
- D. 45

562. Nonlinear functions · Medium · MC

If $f(x) = 4x - 1$ and $g(x) = 4x - 3$, what is the value of $f(g(4))$?

- A. 28
- B. 51
- C. 64
- D. 52

563. Nonlinear functions · Medium · MC

If $f(x) = 2x - 3$ and $g(x) = 4x + 5$, what is the value of $f(g(0))$?

- A. 2
- B. 0
- C. 8
- D. 7

564. Nonlinear functions · Medium · MC

If $f(x) = 5x - 1$ and $g(x) = 4x - 2$, what is the value of $f(g(-1))$?

- A. -31
- B. -20
- C. -30
- D. -12

565. Nonlinear functions · Medium · MC

If $f(x) = 2x + 1$ and $g(x) = 4x - 1$, what is the value of $f(g(5))$?

- A. 39
- B. 38
- C. 30
- D. 40

566. Nonlinear functions · Medium · MC

If $f(x) = 2x + 1$ and $g(x) = 4x + 1$, what is the value of $f(g(4))$?

- A. 35
- B. 26
- C. 32
- D. 36

567. Nonlinear functions · Medium · MC

If $f(x) = 4x + 6$ and $g(x) = 2x + 1$, what is the value of $f(g(5))$?

- A. 40
- B. 51
- C. 50
- D. 37

568. Nonlinear functions · Medium · MC

If $f(x) = 2x$ and $g(x) = 2x - 2$, what is the value of $f(g(-2))$?

- A. -10
- B. -8
- C. -12
- D. -11

569. Nonlinear functions · Medium · MC

If $f(x) = 3x + 3$ and $g(x) = 2x - 1$, what is the value of $f(g(1))$?

- A. 5
- B. 7
- C. 6
- D. 8

570. Nonlinear functions · Medium · MC

If $f(x) = 2x + 3$ and $g(x) = 3x + 2$, what is the value of $f(g(4))$?

- A. 25
- B. 31
- C. 24
- D. 32

571. Nonlinear functions · Medium · MC

If $f(x) = 3x - 3$ and $g(x) = 4x + 5$, what is the value of $f(g(4))$?

- A. 60
- B. 30
- C. 61
- D. 48

572. Nonlinear functions · Medium · MC

If $f(x) = 3x - 2$ and $g(x) = 2x + 3$, what is the value of $f(g(-1))$?

- A. -4
- B. 1
- C. 2
- D. -6

573. Nonlinear functions · Medium · MC

If $f(x) = 3x - 4$ and $g(x) = 3x + 5$, what is the value of $f(g(1))$?

- A. 20
- B. 7
- C. 21
- D. 9

574. Nonlinear functions · Medium · SPR

If $f(x) = 4x - 2$ and $g(x) = 3x + 3$, what is the value of $f(g(0))$?

575. Nonlinear functions · Medium · MC

If $f(x) = 3x + 4$ and $g(x) = 4x - 3$, what is the value of $f(g(3))$?

- A. 36
- B. 31
- C. 32
- D. 22

576. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 2)$. If $f(6) = -5$, what is the value of $g(8)$?

- A. -7
- B. -5
- C. 6
- D. -3

577. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 4)$. If $f(-1) = 9$, what is the value of $g(3)$?

- A. -1
- B. 5
- C. 13
- D. 9

578. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 1)$. If $f(4) = 1$, what is the value of $g(5)$?

- A. 4
- B. 0
- C. 1
- D. 2

579. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 3)$. If $f(3) = 6$, what is the value of $g(6)$?

- A. 3
- B. 9
- C. 12
- D. 6

580. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 2)$. If $f(4) = -1$, what is the value of $g(6)$?

- A. 4
- B. -1
- C. 1
- D. -3

581. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 3)$. If $f(2) = -1$, what is the value of $g(5)$?

- A. -4
- B. 4
- C. 2
- D. -1

582. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 1)$. If $f(2) = 5$, what is the value of $g(3)$?

- A. 2
- B. 6
- C. 5
- D. 4

583. Nonlinear functions · Hard · SPR

The function g is defined by $g(x) = f(x - 4)$. If $f(0) = 7$, what is the value of $g(4)$?

584. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 1)$. If $f(4) = 4$, what is the value of $g(5)$?

- A. 3
- B. 9
- C. 4
- D. 5

585. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 2)$. If $f(5) = 10$, what is the value of $g(7)$?

- A. 5
- B. 12
- C. 10
- D. 8

586. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 1)$. If $f(4) = -4$, what is the value of $g(5)$?

- A. -3
- B. -4
- C. 4
- D. -5

587. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 3)$. If $f(4) = -3$, what is the value of $g(7)$?

- A. 0
- B. -3
- C. -6
- D. 4

588. Nonlinear functions · Hard · SPR

The function g is defined by $g(x) = f(x - 2)$. If $f(4) = 8$, what is the value of $g(6)$?

589. Nonlinear functions · Hard · MC

The function g is defined by $g(x) = f(x - 2)$. If $f(6) = 0$, what is the value of $g(8)$?

- A. 0
- B. -2
- C. 6
- D. 2

590. Nonlinear functions · Hard · SPR

The function g is defined by $g(x) = f(x - 4)$. If $f(1) = 5$, what is the value of $g(5)$?

591. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x + 6} = 7$?

- A. 49
- B. 43
- C. 55
- D. 13

592. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x + 12} = 6$?

- A. 24
- B. 48
- C. 18
- D. 36

593. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x+2} = 4$?

- A. 16
- B. 18
- C. 6
- D. 14

594. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x+1} = 2$?

- A. 3
- B. 4
- C. -1
- D. 5

595. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x+1} = 3$?

- A. 8
- B. 4
- C. 9
- D. 10

596. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x+4} = 4$?

- A. 8
- B. 12
- C. 20
- D. 16

597. Nonlinear equations · Medium · SPR

What is the solution of $\sqrt{x+5} = 9$?

598. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x+8} = 3$?

- A. 9
- B. 17
- C. 11
- D. 1

599. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x+1} = 9$?

- A. 10
- B. 81
- C. 82
- D. 80

600. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x+6} = 5$?

- A. 19
- B. 11
- C. 31
- D. 25

601. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x+2} = 6$?

- A. 34
- B. 8
- C. 38
- D. 36

602. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x+7} = 9$?

- A. 81
- B. 74
- C. 88
- D. 16

603. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x+7} = 7$?

- A. 42
- B. 14
- C. 56
- D. 49

604. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x+2} = 8$?

- A. 62
- B. 66
- C. 64
- D. 10

605. Nonlinear equations · Medium · MC

What is the solution of $\sqrt{x+5} = 3$?

- A. 9
- B. 4
- C. 8
- D. 14

606. Nonlinear equations · Hard · MC

What is the solution of $\sqrt{x+11} = 9$?

- A. 70
- B. 81
- C. 92
- D. 20

607. Nonlinear equations · Hard · MC

What is the solution of $\sqrt{x+4} = 3$?

- A. 13
- B. 7
- C. 5
- D. 9

608. Nonlinear equations · Hard · MC

What is the solution of $\sqrt{x+3} = 8$?

- A. 64
- B. 61
- C. 11
- D. 67

609. Nonlinear equations · Hard · SPR

What is the solution of $\sqrt{x+6} = 6$?

610. Nonlinear equations · Hard · MC

What is the solution of $\sqrt{x+9} = 7$?

- A. 16
- B. 40
- C. 58
- D. 49

611. Nonlinear equations · Hard · SPR

What is the solution of $\sqrt{x+11} = 4$?

612. Nonlinear equations · Hard · MC

What is the solution of $\sqrt{x+3} = 5$?

- A. 8
- B. 22
- C. 25
- D. 28

613. Nonlinear equations · Hard · SPR

What is the solution of $\sqrt{x+11} = 5$?

614. Nonlinear equations · Hard · MC

What is the solution of $\sqrt{x+8} = 9$?

- A. 73
- B. 17
- C. 81
- D. 89

615. Nonlinear equations · Hard · SPR

What is the solution of $\sqrt{x+5} = 7$?

616. Nonlinear equations · Medium · SPR

What value of x satisfies $\frac{x+3}{x+2} = 2$, given that the expression is defined?

617. Nonlinear equations · Medium · SPR

What value of x satisfies $\frac{x+4}{x+1} = 4$, given that the expression is defined?

618. Nonlinear equations · Medium · SPR

What value of x satisfies $\frac{x+8}{x+1} = 2$, given that the expression is defined?

619. Nonlinear equations · Medium · MC

What value of x satisfies $\frac{x+9}{x+4} = 2$, given that the expression is defined?

- A. 10
- B. 3
- C. 1
- D. -4

620. Nonlinear equations · Medium · SPR

What value of x satisfies $\frac{x+20}{x+4} = 3$, given that the expression is defined?

621. Nonlinear equations · Medium · MC

What value of x satisfies $\frac{x+22}{x+4} = 3$, given that the expression is defined?

- A. 5
- B. -4
- C. 27
- D. 8

622. Nonlinear equations · Medium · MC

What value of x satisfies $\frac{x+28}{x+1} = 4$, given that the expression is defined?

- A. -1
- B. 8
- C. 12
- D. 36

623. Nonlinear equations · Medium · SPR

What value of x satisfies $\frac{x+17}{x+6} = 2$, given that the expression is defined?

624. Nonlinear equations · Medium · MC

What value of x satisfies $\frac{x+37}{x+4} = 4$, given that the expression is defined?

- A. 11
- B. 44
- C. -4
- D. 7

625. Nonlinear equations · Medium · MC

What value of x satisfies $\frac{x+15}{x+4} = 2$, given that the expression is defined?

- A. -4
- B. 22
- C. 7
- D. 9

626. Nonlinear equations · Medium · MC

What value of x satisfies $\frac{x+18}{x+2} = 3$, given that the expression is defined?

- A. -2
- B. 9
- C. 24
- D. 6

627. Nonlinear equations · Medium · MC

What value of x satisfies $\frac{x+13}{x+1} = 4$, given that the expression is defined?

- A. 16
- B. -1
- C. 7
- D. 3

628. Nonlinear equations · Medium · MC

What value of x satisfies $\frac{x+4}{x+1} = 2$, given that the expression is defined?

- A. 4
- B. 6
- C. -1
- D. 2

629. Nonlinear equations · Medium · MC

What value of x satisfies $\frac{x+14}{x+2} = 4$, given that the expression is defined?

- A. -2
- B. 6
- C. 16
- D. 2

630. Nonlinear equations · Medium · MC

What value of x satisfies $\frac{x+17}{x+1} = 3$, given that the expression is defined?

- A. 10
- B. 24
- C. -1
- D. 7

631. Nonlinear equations · Hard · SPR

What value of x satisfies $\frac{x+22}{x+6} = 3$, given that the expression is defined?

632. Nonlinear equations · Hard · MC

What value of x satisfies $\frac{x+35}{x+5} = 4$, given that the expression is defined?

- A. 9
- B. -5
- C. 5
- D. 40

633. Nonlinear equations · Hard · SPR

What value of x satisfies $\frac{x+32}{x+2} = 4$, given that the expression is defined?

634. Nonlinear equations · Hard · MC

What value of x satisfies $\frac{x+14}{x+4} = 3$, given that the expression is defined?

A. -4

B. 1

C. 15

D. 4

635. Nonlinear equations · Hard · MC

What value of x satisfies $\frac{x+29}{x+2} = 4$, given that the expression is defined?

A. 11

B. 36

C. -2

D. 7

636. Nonlinear equations · Hard · SPR

What value of x satisfies $\frac{x+6}{x+2} = 2$, given that the expression is defined?

637. Nonlinear equations · Hard · MC

What value of x satisfies $\frac{x+5}{x+3} = 2$, given that the expression is defined?

A. 4

B. -3

C. 1

D. -1

638. Nonlinear equations · Hard · MC

What value of x satisfies $\frac{x+15}{x+5} = 2$, given that the expression is defined?

A. 5

B. 7

C. 20

D. -5

639. Nonlinear equations · Hard · MC

What value of x satisfies $\frac{x+13}{x+4} = 2$, given that the expression is defined?

A. -4

B. 18

C. 7

D. 5

640. Nonlinear equations · Hard · SPR

What value of x satisfies $\frac{x+18}{x+6} = 2$, given that the expression is defined?

641. Nonlinear equations · Hard · SPR

What value of x satisfies $\frac{x+20}{x+6} = 3$, given that the expression is defined?

642. Nonlinear equations · Hard · MC

What value of x satisfies $\frac{x+16}{x+4} = 3$, given that the expression is defined?

- A. 18
- B. 5
- C. 2
- D. -4

643. Nonlinear equations · Hard · MC

What value of x satisfies $\frac{x-1}{x+2} = 2$, given that the expression is defined?

- A. -6
- B. -2
- C. -5
- D. -3

644. Nonlinear equations · Hard · MC

What value of x satisfies $\frac{x+6}{x+5} = 2$, given that the expression is defined?

- A. -4
- B. 2
- C. -5
- D. -2

645. Nonlinear equations · Hard · MC

What value of x satisfies $\frac{x+11}{x+2} = 2$, given that the expression is defined?

- A. -2
- B. 18
- C. 9
- D. 7

646. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 11x$ intersect at the origin and one other point. What is the positive x -coordinate of that other point?

- A. 12
- B. 121
- C. 0
- D. 11

647. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 2x$ intersect at the origin and one other point. What is the positive x -coordinate of that other point?

- A. 3
- B. 4
- C. 0
- D. 2

648. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 16x$ intersect at the origin and one other point. What is the positive x -coordinate of that other point?

- A. 0
- B. 256
- C. 16
- D. 17

649. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 4x$ intersect at the origin and one other point. What is the positive x -coordinate of that other point?

- A. 5
- B. 16
- C. 4
- D. 0

650. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 13x$ intersect at the origin and one other point. What is the positive x -coordinate of that other point?

- A. 169
- B. 0
- C. 13
- D. 14

651. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 9x$ intersect at the origin and one other point. What is the positive x -coordinate of that other point?

- A. 10
- B. 9
- C. 0
- D. 81

652. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 7x$ intersect at the origin and one other point. What is the positive x -coordinate of that other point?

- A. 0
- B. 8
- C. 49
- D. 7

653. Systems of equations · Medium · SPR

The graphs of $y = x^2$ and $y = 3x$ intersect at the origin and one other point. What is the positive x -coordinate of that other point?

654. Systems of equations · Medium · SPR

The graphs of $y = x^2$ and $y = 6x$ intersect at the origin and one other point. What is the positive x -coordinate of that other point?

655. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 5x$ intersect at the origin and one other point. What is the positive x -coordinate of that other point?

- A. 5
- B. 0
- C. 25
- D. 6

656. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 8x$ intersect at the origin and one other point. What is the positive x -coordinate of that other point?

- A. 64
- B. 8
- C. 0
- D. 9

657. Systems of equations · Medium · MC

The graphs of $y = x^2$ and $y = 10x$ intersect at the origin and one other point. What is the positive x -coordinate of that other point?

- A. 100
- B. 11
- C. 10
- D. 0

658. Systems of equations · Hard · SPR

The graphs of $y = x^2$ and $y = 9x$ intersect at the origin and one other point. What is the y -coordinate of that other point?

659. Systems of equations · Hard · SPR

The graphs of $y = x^2$ and $y = 11x$ intersect at the origin and one other point. What is the y -coordinate of that other point?

660. Systems of equations · Hard · MC

The graphs of $y = x^2$ and $y = 6x$ intersect at the origin and one other point. What is the y -coordinate of that other point?

- A. 36
- B. 12
- C. 216
- D. 6

661. Systems of equations · Hard · MC

The graphs of $y = x^2$ and $y = 2x$ intersect at the origin and one other point. What is the y -coordinate of that other point?

- A. 2
- B. 0
- C. 8
- D. 4

662. Systems of equations · Hard · MC

The graphs of $y = x^2$ and $y = 13x$ intersect at the origin and one other point. What is the y -coordinate of that other point?

- A. 169
- B. 26
- C. 13
- D. 2197

663. Systems of equations · Hard · MC

The graphs of $y = x^2$ and $y = 10x$ intersect at the origin and one other point. What is the y -coordinate of that other point?

- A. 20
- B. 100
- C. 1000
- D. 10

664. Systems of equations · Hard · MC

The graphs of $y = x^2$ and $y = 8x$ intersect at the origin and one other point. What is the y -coordinate of that other point?

- A. 16
- B. 512
- C. 64
- D. 8

665. Systems of equations · Hard · MC

The graphs of $y = x^2$ and $y = 16x$ intersect at the origin and one other point. What is the y -coordinate of that other point?

- A. 32
- B. 4096
- C. 16
- D. 256

666. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 - 24x + 135 = 0$ have?

- A. Exactly two
- B. Infinitely many
- C. None
- D. Exactly one

667. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 + 4x + 4 = 0$ have?

- A. Exactly two
- B. Infinitely many
- C. None
- D. Exactly one

668. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 - 13x + 40 = 0$ have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. None

669. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 + 6x + 15 = 0$ have?

- A. None
- B. Exactly one
- C. Exactly two
- D. Infinitely many

670. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 - 22x + 117 = 0$ have?

- A. None
- B. Exactly one
- C. Exactly two
- D. Infinitely many

671. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 + 6x + 17 = 0$ have?

- A. Exactly two
- B. Infinitely many
- C. None
- D. Exactly one

672. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 + 8x + 21 = 0$ have?

- A. None
- B. Exactly one
- C. Exactly two
- D. Infinitely many

673. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 + 2x + 11 = 0$ have?

- A. None
- B. Exactly one
- C. Exactly two
- D. Infinitely many

674. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 + 14x + 49 = 0$ have?

- A. None
- B. Exactly one
- C. Exactly two
- D. Infinitely many

675. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 - 4x + 4 = 0$ have?

- A. Infinitely many
- B. None
- C. Exactly one
- D. Exactly two

676. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 + 8 = 0$ have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. None

677. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 + 6x + 9 = 0$ have?

- A. Infinitely many
- B. None
- C. Exactly one
- D. Exactly two

678. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 + 1x + 8 = 0$ have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. None

679. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 + 3x + 4 = 0$ have?

- A. None
- B. Exactly one
- C. Exactly two
- D. Infinitely many

680. Nonlinear equations · Medium · MC

How many distinct real solutions does $x^2 + 7x + 16 = 0$ have?

- A. None
- B. Exactly one
- C. Exactly two
- D. Infinitely many

681. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 4)(x + 5) - 2(x - 4)$?

- A. $(x - 4)(x + 3)$
- B. $(x - 4)(x + 7)$
- C. $(x + 4)(x + 3)$
- D. $(x - 4)(x - 3)$

682. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 3)(x + 4) - 4(x - 3)$?

- A. $(x - 3)(x + 0)$
- B. $(x - 3)(x + 8)$
- C. $(x + 3)(x + 0)$
- D. $(x - 3)(x - 0)$

683. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 3)(x + 3) - 3(x - 3)$?

- A. $(x - 3)(x + 0)$
- B. $(x - 3)(x + 6)$
- C. $(x + 3)(x + 0)$
- D. $(x - 3)(x - 0)$

684. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 6)(x + 7) - 5(x - 6)$?

- A. $(x - 6)(x + 2)$
- B. $(x - 6)(x + 12)$
- C. $(x + 6)(x + 2)$
- D. $(x - 6)(x - 2)$

685. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 2)(x + 2) - 4(x - 2)$?

- A. $(x + 2)(x + -2)$
- B. $(x - 2)(x - -2)$
- C. $(x - 2)(x + -2)$
- D. $(x - 2)(x + 6)$

686. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 5)(x + 3) - 3(x - 5)$?

- A. $(x - 5)(x + 0)$
- B. $(x - 5)(x + 6)$
- C. $(x + 5)(x + 0)$
- D. $(x - 5)(x - 0)$

687. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 2)(x + 6) - 2(x - 2)$?

- A. $(x + 2)(x + 4)$
- B. $(x - 2)(x - 4)$
- C. $(x - 2)(x + 4)$
- D. $(x - 2)(x + 8)$

688. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 5)(x + 7) - 4(x - 5)$?

- A. $(x - 5)(x + 11)$
- B. $(x + 5)(x + 3)$
- C. $(x - 5)(x - 3)$
- D. $(x - 5)(x + 3)$

689. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 6)(x + 6) - 3(x - 6)$?

- A. $(x - 6)(x - 3)$
- B. $(x - 6)(x + 3)$
- C. $(x - 6)(x + 9)$
- D. $(x + 6)(x + 3)$

690. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 2)(x + 5) - 3(x - 2)$?

- A. $(x - 2)(x - 2)$
- B. $(x - 2)(x + 2)$
- C. $(x - 2)(x + 8)$
- D. $(x + 2)(x + 2)$

691. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 2)(x + 4) - 4(x - 2)$?

- A. $(x + 2)(x + 0)$
- B. $(x - 2)(x - 0)$
- C. $(x - 2)(x + 0)$
- D. $(x - 2)(x + 8)$

692. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 4)(x + 6) - 5(x - 4)$?

- A. $(x + 4)(x + 1)$
- B. $(x - 4)(x - 1)$
- C. $(x - 4)(x + 1)$
- D. $(x - 4)(x + 11)$

693. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 5)(x + 4) - 5(x - 5)$?

- A. $(x + 5)(x + -1)$
- B. $(x - 5)(x - -1)$
- C. $(x - 5)(x + -1)$
- D. $(x - 5)(x + 9)$

694. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 6)(x + 2) - 2(x - 6)$?

- A. $(x - 6)(x + 4)$
- B. $(x + 6)(x + 0)$
- C. $(x - 6)(x - 0)$
- D. $(x - 6)(x + 0)$

695. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 6)(x + 7) - 3(x - 6)$?

- A. $(x - 6)(x + 10)$
- B. $(x + 6)(x + 4)$
- C. $(x - 6)(x - 4)$
- D. $(x - 6)(x + 4)$

696. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 3)(x + 7) - 2(x - 3)$?

- A. $(x - 3)(x + 9)$
- B. $(x + 3)(x + 5)$
- C. $(x - 3)(x - 5)$
- D. $(x - 3)(x + 5)$

697. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 3)(x + 4) - 3(x - 3)$?

- A. $(x - 3)(x + 1)$
- B. $(x - 3)(x + 7)$
- C. $(x + 3)(x + 1)$
- D. $(x - 3)(x - 1)$

698. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 5)(x + 4) - 4(x - 5)$?

- A. $(x - 5)(x - 0)$
- B. $(x - 5)(x + 0)$
- C. $(x - 5)(x + 8)$
- D. $(x + 5)(x + 0)$

699. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 4)(x + 2) - 3(x - 4)$?

- A. $(x - 4)(x + 5)$
- B. $(x + 4)(x + -1)$
- C. $(x - 4)(x - -1)$
- D. $(x - 4)(x + -1)$

700. Equivalent expressions · Hard · MC

Which expression is equivalent to $(x - 2)(x + 7) - 5(x - 2)$?

- A. $(x - 2)(x - 2)$
- B. $(x - 2)(x + 2)$
- C. $(x - 2)(x + 12)$
- D. $(x + 2)(x + 2)$

Problem-Solving and Data Analysis

701. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 3 : 2. There are 12 red marbles. How many marbles are in the bag?

- A. 24
- B. 60
- C. 14
- D. 20

702. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 3 : 3. There are 6 red marbles. How many marbles are in the bag?

- A. 9
- B. 36
- C. 12
- D. 18

703. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 3 : 2. There are 6 red marbles. How many marbles are in the bag?

- A. 30
- B. 12
- C. 8
- D. 10

704. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 2 : 3. There are 6 red marbles. How many marbles are in the bag?

- A. 15
- B. 18
- C. 30
- D. 9

705. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 4 : 6. There are 16 red marbles. How many marbles are in the bag?

- A. 40
- B. 96
- C. 22
- D. 160

706. Ratios and rates · Easy · SPR

The ratio of red marbles to blue marbles in a bag is 4 : 5. There are 16 red marbles. How many marbles are in the bag?

707. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 3 : 5. There are 12 red marbles. How many marbles are in the bag?

- A. 60
- B. 96
- C. 32
- D. 17

708. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 2 : 6. There are 4 red marbles. How many marbles are in the bag?

- A. 10
- B. 32
- C. 24
- D. 16

709. Ratios and rates · Easy · MC

The ratio of red marbles to blue marbles in a bag is 4 : 2. There are 12 red marbles. How many marbles are in the bag?

- A. 18
- B. 14
- C. 24
- D. 72

710. Ratios and rates · Easy · SPR

The ratio of red marbles to blue marbles in a bag is 5 : 6. There are 20 red marbles. How many marbles are in the bag?

711. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 3 : 3. There are 18 red marbles. How many marbles are in the bag?

712. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 4 : 3. There are 20 red marbles. How many marbles are in the bag?

713. Ratios and rates · Medium · MC

The ratio of red marbles to blue marbles in a bag is 3 : 2. There are 18 red marbles. How many marbles are in the bag?

- A. 36
- B. 90
- C. 30
- D. 20

714. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 4 : 6. There are 12 red marbles. How many marbles are in the bag?

715. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 4 : 5. There are 24 red marbles. How many marbles are in the bag?

716. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 4 : 2. There are 24 red marbles. How many marbles are in the bag?

717. Ratios and rates · Medium · MC

The ratio of red marbles to blue marbles in a bag is 5 : 2. There are 40 red marbles. How many marbles are in the bag?

- A. 80
- B. 280
- C. 56
- D. 42

718. Ratios and rates · Medium · MC

The ratio of red marbles to blue marbles in a bag is 4 : 3. There are 24 red marbles. How many marbles are in the bag?

- A. 168
- B. 27
- C. 72
- D. 42

719. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 5 : 5. There are 25 red marbles. How many marbles are in the bag?

720. Ratios and rates · Medium · SPR

The ratio of red marbles to blue marbles in a bag is 2 : 6. There are 8 red marbles. How many marbles are in the bag?

721. Ratios and rates · Hard · MC

The ratio of red marbles to blue marbles in a bag is 3 : 3. There are 9 red marbles. How many marbles are in the bag?

- A. 54
- B. 27
- C. 12
- D. 18

722. Ratios and rates · Hard · SPR

The ratio of red marbles to blue marbles in a bag is 4 : 2. There are 16 red marbles. How many marbles are in the bag?

723. Ratios and rates · Hard · SPR

The ratio of red marbles to blue marbles in a bag is 5 : 3. There are 40 red marbles. How many marbles are in the bag?

724. Ratios and rates · Hard · MC

The ratio of red marbles to blue marbles in a bag is 4 : 5. There are 20 red marbles. How many marbles are in the bag?

- A. 180
- B. 100
- C. 45
- D. 25

725. Ratios and rates · Hard · MC

The ratio of red marbles to blue marbles in a bag is 5 : 3. There are 20 red marbles. How many marbles are in the bag?

- A. 32
- B. 160
- C. 60
- D. 23

726. Ratios and rates · Easy · MC

A car travels 210 miles in 5 hours at a constant speed. What is the speed in miles per hour?

- A. 42
- B. 5
- C. 215
- D. 1050

727. Ratios and rates · Easy · MC

A car travels 150 miles in 2 hours at a constant speed. What is the speed in miles per hour?

- A. 152
- B. 300
- C. 75
- D. 2

728. Ratios and rates · Easy · MC

A car travels 180 miles in 2 hours at a constant speed. What is the speed in miles per hour?

- A. 2
- B. 182
- C. 360
- D. 90

729. Ratios and rates · Easy · SPR

A car travels 300 miles in 2 hours at a constant speed. What is the speed in miles per hour?

730. Ratios and rates · Easy · MC

A car travels 240 miles in 2 hours at a constant speed. What is the speed in miles per hour?

- A. 480
- B. 2
- C. 242
- D. 120

731. Ratios and rates · Easy · SPR

A car travels 150 miles in 5 hours at a constant speed. What is the speed in miles per hour?

732. Ratios and rates · Easy · MC

A car travels 240 miles in 3 hours at a constant speed. What is the speed in miles per hour?

- A. 3
- B. 243
- C. 80
- D. 720

733. Ratios and rates · Easy · MC

A car travels 240 miles in 5 hours at a constant speed. What is the speed in miles per hour?

- A. 1200
- B. 5
- C. 48
- D. 245

734. Percentages · Easy · MC

A jacket costs \$80. During a sale the price is reduced by 30%. What is the sale price, in dollars?

- A. 104
- B. 24
- C. 2400
- D. 56

735. Percentages · Easy · MC

A jacket costs \$100. During a sale the price is reduced by 10%. What is the sale price, in dollars?

- A. 1000
- B. 90
- C. 10
- D. 110

736. Percentages · Easy · SPR

A jacket costs \$200. During a sale the price is reduced by 20%. What is the sale price, in dollars?

737. Percentages · Easy · MC

A jacket costs \$40. During a sale the price is reduced by 25%. What is the sale price, in dollars?

- A. 30
- B. 1000
- C. 50
- D. 10

738. Percentages · Easy · MC

A jacket costs \$60. During a sale the price is reduced by 20%. What is the sale price, in dollars?

- A. 12
- B. 1200
- C. 72
- D. 48

739. Percentages · Easy · MC

A jacket costs \$120. During a sale the price is reduced by 10%. What is the sale price, in dollars?

- A. 12
- B. 1200
- C. 108
- D. 132

740. Percentages · Easy · MC

A jacket costs \$50. During a sale the price is reduced by 15%. What is the sale price, in dollars?

- A. 7
- B. 43
- C. 57
- D. 750

741. Percentages · Easy · MC

A jacket costs \$50. During a sale the price is reduced by 30%. What is the sale price, in dollars?

- A. 35
- B. 15
- C. 1500
- D. 65

742. Percentages · Easy · SPR

A jacket costs \$120. During a sale the price is reduced by 15%. What is the sale price, in dollars?

743. Percentages · Easy · MC

A jacket costs \$100. During a sale the price is reduced by 30%. What is the sale price, in dollars?

- A. 3000
- B. 130
- C. 30
- D. 70

744. Percentages · Easy · MC

A jacket costs \$60. During a sale the price is reduced by 15%. What is the sale price, in dollars?

- A. 51
- B. 900
- C. 69
- D. 9

745. Percentages · Easy · SPR

A jacket costs \$60. During a sale the price is reduced by 10%. What is the sale price, in dollars?

746. Percentages · Medium · MC

A bike is listed at \$100. It is discounted by 25%, and then a 8% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 81
- B. 67
- C. 83
- D. 75

747. Percentages · Medium · SPR

A bike is listed at \$200. It is discounted by 25%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

748. Percentages · Medium · SPR

A bike is listed at \$50. It is discounted by 20%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

749. Percentages · Medium · MC

A bike is listed at \$80. It is discounted by 25%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 68
- B. 60
- C. 66
- D. 45

750. Percentages · Medium · MC

A bike is listed at \$50. It is discounted by 20%, and then a 5% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 45
- B. 25
- C. 42
- D. 40

751. Percentages · Medium · MC

A bike is listed at \$200. It is discounted by 30%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 150
- B. 140
- C. 160
- D. 154

752. Percentages · Medium · MC

A bike is listed at \$200. It is discounted by 25%, and then a 8% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 167
- B. 150
- C. 162
- D. 166

753. Percentages · Medium · SPR

A bike is listed at \$200. It is discounted by 20%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

754. Percentages · Medium · MC

A bike is listed at \$40. It is discounted by 25%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 34
- B. 5
- C. 30
- D. 33

755. Percentages · Medium · MC

A bike is listed at \$200. It is discounted by 10%, and then a 5% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 190
- B. 180
- C. 185
- D. 189

756. Percentages · Hard · SPR

A bike is listed at \$200. It is discounted by 30%, and then a 5% sales tax is added to the discounted price. What is the final price, in dollars?

757. Percentages · Hard · MC

A bike is listed at \$120. It is discounted by 25%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 99
- B. 102
- C. 90
- D. 85

758. Percentages · Hard · MC

A bike is listed at \$100. It is discounted by 20%, and then a 5% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 85
- B. 75
- C. 80
- D. 84

759. Percentages · Hard · MC

A bike is listed at \$80. It is discounted by 25%, and then a 5% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 50
- B. 63
- C. 60
- D. 64

760. Percentages · Hard · MC

A bike is listed at \$200. It is discounted by 15%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 187
- B. 170
- C. 190
- D. 175

761. Percentages · Hard · MC

A bike is listed at \$100. It is discounted by 20%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 70
- B. 80
- C. 88
- D. 90

762. Percentages · Hard · MC

A bike is listed at \$100. It is discounted by 30%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 80
- B. 77
- C. 60
- D. 70

763. Percentages · Hard · MC

A bike is listed at \$100. It is discounted by 10%, and then a 10% sales tax is added to the discounted price. What is the final price, in dollars?

- A. 100
- B. 80
- C. 99
- D. 90

764. One-variable data · Easy · MC

The mean of the numbers 6, 17, 6, 8, and x is 9. What is the value of x ?

- A. 45
- B. 9
- C. 7
- D. 8

765. One-variable data · Easy · MC

The mean of the numbers 10, 12, 8, 11, and x is 12. What is the value of x ?

- A. 12
- B. 20
- C. 10
- D. 19

766. One-variable data · Easy · MC

The mean of the numbers 19, 9, 11, 5, and x is 15. What is the value of x ?

- A. 11
- B. 31
- C. 32
- D. 15

767. One-variable data · Easy · MC

The mean of the numbers 18, 4, 15, 10, and x is 10. What is the value of x ?

- A. 3
- B. 11
- C. 10
- D. 4

768. One-variable data · Easy · SPR

The mean of the numbers 11, 9, 17, 18, and x is 14. What is the value of x ?

769. One-variable data · Easy · SPR

The mean of the numbers 10, 20, 20, 14, and x is 15. What is the value of x ?

770. One-variable data · Medium · MC

The mean of the numbers 12, 6, 16, 10, and x is 12. What is the value of x ?

- A. 16
- B. 17
- C. 11
- D. 12

771. One-variable data · Medium · MC

The mean of the numbers 8, 10, 18, 8, and x is 16. What is the value of x ?

- A. 16
- B. 11
- C. 36
- D. 37

772. One-variable data · Medium · MC

The mean of the numbers 8, 14, 5, 14, and x is 11. What is the value of x ?

- A. 11
- B. 15
- C. 10
- D. 14

773. One-variable data · Medium · MC

The mean of the numbers 4, 6, 14, 17, and x is 16. What is the value of x ?

- A. 39
- B. 40
- C. 10
- D. 16

774. One-variable data · Medium · MC

The mean of the numbers 16, 6, 14, 20, and x is 13. What is the value of x ?

- A. 9
- B. 14
- C. 13
- D. 10

775. One-variable data · Medium · MC

The mean of the numbers 10, 12, 8, 11, and x is 10. What is the value of x ?

- A. 50
- B. 8
- C. 10
- D. 9

776. One-variable data · Easy · MC

What is the median of the data set 5, 11, 16, 19, 26?

- A. 15
- B. 26
- C. 5
- D. 16

777. One-variable data · Easy · MC

What is the median of the data set 12, 17, 22, 24, 25?

- A. 22
- B. 20
- C. 12
- D. 25

778. One-variable data · Easy · MC

What is the median of the data set 6, 11, 13, 14, 26?

- A. 14
- B. 13
- C. 6
- D. 26

779. One-variable data · Easy · MC

What is the median of the data set 10, 12, 20, 24, 29?

- A. 19
- B. 29
- C. 10
- D. 20

780. One-variable data · Easy · MC

What is the median of the data set 6, 13, 14, 17, 24?

- A. 13
- B. 6
- C. 24
- D. 14

781. One-variable data · Easy · MC

What is the median of the data set 3, 5, 9, 10, 11?

- A. 9
- B. 11
- C. 3
- D. 7

782. One-variable data · Easy · MC

What is the median of the data set 7, 16, 17, 22, 23?

- A. 7
- B. 23
- C. 17
- D. 16

783. One-variable data · Easy · MC

What is the median of the data set 5, 6, 12, 16, 17?

- A. 17
- B. 12
- C. 11
- D. 5

784. One-variable data · Medium · MC

What is the median of the data set 4, 11, 23, 25, 26?

- A. 17
- B. 4
- C. 23
- D. 26

785. One-variable data · Medium · MC

What is the median of the data set 13, 18, 19, 21, 23?

- A. 13
- B. 23
- C. 18
- D. 19

786. One-variable data · Medium · MC

What is the median of the data set 18, 21, 22, 23, 29?

- A. 22
- B. 18
- C. 29
- D. 21

787. One-variable data · Medium · MC

What is the median of the data set 11, 13, 18, 19, 25?

- A. 11
- B. 17
- C. 25
- D. 18

788. One-variable data · Medium · MC

What is the median of the data set 11, 12, 17, 19, 22?

- A. 17
- B. 11
- C. 22
- D. 16

789. One-variable data · Medium · MC

What is the median of the data set 3, 14, 15, 18, 27?

- A. 15
- B. 3
- C. 14
- D. 27

790. Probability · Easy · MC

A bag has 6 red, 2 blue, and 4 green counters. One counter is drawn at random. What is the probability that it is red?

- A. $\frac{1}{3}$
- B. $\frac{1}{2}$
- C. $\frac{1}{6}$
- D. 3

791. Probability · Easy · MC

A bag has 7 red, 7 blue, and 4 green counters. One counter is drawn at random. What is the probability that it is red?

- A. $\frac{1}{3}$
- B. $\frac{7}{18}$
- C. 1
- D. $\frac{25}{18}$

792. Probability · Easy · MC

A bag has 3 red, 3 blue, and 4 green counters. One counter is drawn at random. What is the probability that it is red?

- A. $\frac{1}{3}$
- B. $\frac{13}{10}$
- C. $\frac{3}{10}$
- D. 1

793. Probability · Easy · MC

A bag has 5 red, 3 blue, and 1 green counters. One counter is drawn at random. What is the probability that it is red?

- A. $\frac{5}{9}$
- B. $\frac{14}{9}$
- C. $\frac{5}{3}$
- D. $\frac{1}{3}$

794. Probability · Easy · SPR

A bag has 7 red, 2 blue, and 4 green counters. One counter is drawn at random. What is the probability that it is red?

795. Probability · Easy · MC

A bag has 3 red, 2 blue, and 3 green counters. One counter is drawn at random. What is the probability that it is red?

- A. $\frac{1}{3}$
- B. $\frac{3}{2}$
- C. $\frac{3}{8}$
- D. $\frac{1}{4}$

796. Probability · Easy · MC

A bag has 4 red, 4 blue, and 5 green counters. One counter is drawn at random. What is the probability that it is red?

- A. $\frac{1}{3}$
- B. 1
- C. $\frac{17}{13}$
- D. $\frac{4}{13}$

797. Probability · Easy · MC

A bag has 8 red, 4 blue, and 5 green counters. One counter is drawn at random. What is the probability that it is red?

- A. $\frac{1}{3}$
- B. $\frac{4}{17}$
- C. $\frac{8}{17}$
- D. 2

798. Probability · Medium · SPR

A bag has 7 red, 3 blue, and 2 green counters. One counter is drawn at random. What is the probability that it is red?

799. Probability · Medium · SPR

A bag has 3 red, 4 blue, and 6 green counters. One counter is drawn at random. What is the probability that it is red?

800. Probability · Medium · MC

A bag has 7 red, 7 blue, and 6 green counters. One counter is drawn at random. What is the probability that it is red?

- A. 1
- B. $\frac{1}{3}$
- C. $\frac{27}{20}$
- D. $\frac{7}{20}$

801. Probability · Medium · MC

A bag has 4 red, 2 blue, and 3 green counters. One counter is drawn at random. What is the probability that it is red?

- A. $\frac{1}{3}$
- B. $\frac{2}{9}$
- C. 2
- D. $\frac{4}{9}$

802. Probability · Medium · MC

A bag has 4 red, 7 blue, and 2 green counters. One counter is drawn at random. What is the probability that it is red?

- A. $\frac{4}{7}$
- B. $\frac{4}{13}$
- C. $\frac{1}{3}$
- D. $\frac{7}{13}$

803. Probability · Medium · MC

A bag has 7 red, 3 blue, and 6 green counters. One counter is drawn at random. What is the probability that it is red?

- A. $\frac{1}{3}$
- B. $\frac{7}{3}$
- C. $\frac{3}{16}$
- D. $\frac{7}{16}$

804. Probability · Hard · SPR

A bag has 5 red, 5 blue, and 6 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

805. Probability · Hard · MC

A bag has 7 red, 5 blue, and 1 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

- A. $\frac{7}{13}$
- B. $\frac{1}{13}$
- C. $\frac{7}{12}$
- D. $\frac{5}{12}$

806. Probability · Hard · SPR

A bag has 9 red, 6 blue, and 6 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

807. Probability · Hard · MC

A bag has 7 red, 2 blue, and 1 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

- A. $\frac{2}{9}$
- B. $\frac{1}{10}$
- C. $\frac{7}{10}$
- D. $\frac{7}{9}$

808. Probability · Hard · MC

A bag has 7 red, 6 blue, and 6 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

- A. $\frac{6}{19}$
- B. $\frac{6}{13}$
- C. $\frac{7}{19}$
- D. $\frac{7}{13}$

809. Probability · Hard · MC

A bag has 6 red, 8 blue, and 3 green counters. One counter is drawn at random. Given that it is not green, what is the probability that it is red?

- A. $\frac{6}{17}$
- B. $\frac{3}{17}$
- C. $\frac{3}{7}$
- D. $\frac{4}{7}$

810. Probability · Medium · MC

A school recorded whether students joined a club. Among juniors, 10 joined and 12 did not. Among seniors, 8 joined and 7 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

- A. $\frac{5}{9}$
- B. $\frac{4}{9}$
- C. $\frac{5}{11}$
- D. $\frac{10}{37}$

811. Probability · Medium · MC

A school recorded whether students joined a club. Among juniors, 11 joined and 15 did not. Among seniors, 7 joined and 15 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

- A. $\frac{11}{26}$
- B. $\frac{11}{18}$
- C. $\frac{11}{48}$
- D. $\frac{7}{18}$

812. Probability · Medium · MC

A school recorded whether students joined a club. Among juniors, 18 joined and 15 did not. Among seniors, 10 joined and 8 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

- A. $\frac{6}{17}$
- B. $\frac{9}{14}$
- C. $\frac{5}{14}$
- D. $\frac{6}{11}$

813. Probability · Medium · SPR

A school recorded whether students joined a club. Among juniors, 18 joined and 10 did not. Among seniors, 12 joined and 8 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

814. Probability · Medium · MC

A school recorded whether students joined a club. Among juniors, 11 joined and 14 did not. Among seniors, 12 joined and 14 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

- A. $\frac{11}{23}$
- B. $\frac{11}{51}$
- C. $\frac{12}{23}$
- D. $\frac{11}{25}$

815. Probability · Medium · MC

A school recorded whether students joined a club. Among juniors, 13 joined and 7 did not. Among seniors, 7 joined and 11 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

- A. $\frac{7}{20}$
- B. $\frac{13}{20}$
- C. $\frac{13}{38}$
- D. $\frac{33}{20}$

816. Probability · Medium · MC

A school recorded whether students joined a club. Among juniors, 18 joined and 7 did not. Among seniors, 5 joined and 8 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

- A. $\frac{18}{23}$
- B. $\frac{9}{19}$
- C. $\frac{18}{25}$
- D. $\frac{5}{23}$

817. Probability · Medium · SPR

A school recorded whether students joined a club. Among juniors, 8 joined and 15 did not. Among seniors, 6 joined and 13 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

818. Probability · Hard · MC

A school recorded whether students joined a club. Among juniors, 10 joined and 13 did not. Among seniors, 10 joined and 11 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

- A. $\frac{10}{23}$
- B. $\frac{3}{2}$
- C. $\frac{1}{2}$
- D. $\frac{5}{22}$

819. Probability · Hard · SPR

A school recorded whether students joined a club. Among juniors, 17 joined and 6 did not. Among seniors, 9 joined and 12 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

820. Probability · Hard · MC

A school recorded whether students joined a club. Among juniors, 8 joined and 8 did not. Among seniors, 9 joined and 7 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

- A. $\frac{9}{17}$
- B. $\frac{1}{2}$
- C. $\frac{1}{4}$
- D. $\frac{8}{17}$

821. Probability · Hard · MC

A school recorded whether students joined a club. Among juniors, 10 joined and 15 did not. Among seniors, 12 joined and 12 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

- A. $\frac{2}{5}$
- B. $\frac{10}{49}$
- C. $\frac{6}{11}$
- D. $\frac{5}{11}$

822. Probability · Hard · MC

A school recorded whether students joined a club. Among juniors, 12 joined and 14 did not. Among seniors, 12 joined and 9 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

- A. $\frac{3}{2}$
- B. $\frac{6}{13}$
- C. $\frac{1}{2}$
- D. $\frac{12}{47}$

823. Probability · Hard · MC

A school recorded whether students joined a club. Among juniors, 12 joined and 14 did not. Among seniors, 5 joined and 8 did not. A student is chosen at random from those who joined the club. What is the probability that the student is a junior?

- A. $\frac{6}{13}$
- B. $\frac{5}{17}$
- C. $\frac{12}{17}$
- D. $\frac{4}{13}$

824. Two-variable data · Easy · MC

A line of best fit for a data set is $y = 4x + 12$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 9$?

- A. 24
- B. 48
- C. 25
- D. 12

825. Two-variable data · Easy · MC

A line of best fit for a data set is $y = -2x + 8$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 3$?

- A. -14
- B. 2
- C. 9
- D. 8

826. Two-variable data · Easy · MC

A line of best fit for a data set is $y = 3x + 7$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 10$?

- A. 20
- B. 7
- C. 37
- D. 23

827. Two-variable data · Easy · MC

A line of best fit for a data set is $y = 2x + 8$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 5$?

- A. 15
- B. 2
- C. 8
- D. 18

828. Two-variable data · Easy · MC

A line of best fit for a data set is $y = -2x + 10$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 6$?

- A. 10
- B. -2
- C. -22
- D. 14

829. Two-variable data · Easy · SPR

A line of best fit for a data set is $y = -3x + 10$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 7$?

830. Two-variable data · Medium · MC

A line of best fit for a data set is $y = 3x + 11$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 3$?

- A. -2
- B. 20
- C. 17
- D. 11

831. Two-variable data · Medium · MC

A line of best fit for a data set is $y = -2x + 5$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 10$?

- A. -25
- B. 13
- C. -15
- D. 5

832. Two-variable data · Medium · MC

A line of best fit for a data set is $y = 3x + 5$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 6$?

- A. 13
- B. 5
- C. 14
- D. 23

833. Two-variable data · Medium · MC

A line of best fit for a data set is $y = 2x + 4$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 7$?

- A. 18
- B. 10
- C. 13
- D. 4

834. Two-variable data · Medium · MC

A line of best fit for a data set is $y = 3x + 2$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 7$?

- A. 23
- B. 19
- C. 12
- D. 2

835. Two-variable data · Medium · SPR

A line of best fit for a data set is $y = 2x + 12$, where x is the number of hours studied and y is the predicted score. According to the model, what is the predicted score when $x = 3$?

836. Inference from samples · Medium · MC

In a random sample of 600 voters, 52% supported a proposal. The margin of error for this estimate is 3 percentage points. Which conclusion is best supported?

- A. Exactly 52% of all voters support the proposal.
- B. The margin of error would stay 3 percentage points if the sample size were much smaller.
- C. The result proves that the proposal will pass.
- D. It is plausible that between 49% and 55% of all voters support the proposal.

837. Inference from samples · Medium · MC

In a random sample of 900 voters, 54% supported a proposal. The margin of error for this estimate is 5 percentage points. Which conclusion is best supported?

- A. It is plausible that between 49% and 59% of all voters support the proposal.
- B. Exactly 54% of all voters support the proposal.
- C. The margin of error would stay 5 percentage points if the sample size were much smaller.
- D. The result proves that the proposal will pass.

838. Inference from samples · Medium · MC

In a random sample of 400 voters, 57% supported a proposal. The margin of error for this estimate is 5 percentage points. Which conclusion is best supported?

- A. The margin of error would stay 5 percentage points if the sample size were much smaller.
- B. The result proves that the proposal will pass.
- C. It is plausible that between 52% and 62% of all voters support the proposal.
- D. Exactly 57% of all voters support the proposal.

839. Inference from samples · Medium · MC

In a random sample of 900 voters, 48% supported a proposal. The margin of error for this estimate is 5 percentage points. Which conclusion is best supported?

- A. The result proves that the proposal will pass.
- B. It is plausible that between 43% and 53% of all voters support the proposal.
- C. Exactly 48% of all voters support the proposal.
- D. The margin of error would stay 5 percentage points if the sample size were much smaller.

840. Inference from samples · Medium · MC

In a random sample of 1200 voters, 54% supported a proposal. The margin of error for this estimate is 5 percentage points. Which conclusion is best supported?

- A. The margin of error would stay 5 percentage points if the sample size were much smaller.
- B. The result proves that the proposal will pass.
- C. It is plausible that between 49% and 59% of all voters support the proposal.
- D. Exactly 54% of all voters support the proposal.

841. Inference from samples · Medium · MC

In a random sample of 900 voters, 61% supported a proposal. The margin of error for this estimate is 3 percentage points. Which conclusion is best supported?

- A. The margin of error would stay 3 percentage points if the sample size were much smaller.
- B. The result proves that the proposal will pass.
- C. It is plausible that between 58% and 64% of all voters support the proposal.
- D. Exactly 61% of all voters support the proposal.

842. Inference from samples · Hard · MC

In a random sample of 800 voters, 54% supported a proposal. The margin of error for this estimate is 4 percentage points. Which conclusion is best supported?

- A. Exactly 54% of all voters support the proposal.
- B. The margin of error would stay 4 percentage points if the sample size were much smaller.
- C. The result proves that the proposal will pass.
- D. It is plausible that between 50% and 58% of all voters support the proposal.

843. Inference from samples · Hard · MC

In a random sample of 400 voters, 63% supported a proposal. The margin of error for this estimate is 3 percentage points. Which conclusion is best supported?

- A. It is plausible that between 60% and 66% of all voters support the proposal.
- B. Exactly 63% of all voters support the proposal.
- C. The margin of error would stay 3 percentage points if the sample size were much smaller.
- D. The result proves that the proposal will pass.

844. Inference from samples · Hard · MC

In a random sample of 400 voters, 63% supported a proposal. The margin of error for this estimate is 5 percentage points. Which conclusion is best supported?

- A. It is plausible that between 58% and 68% of all voters support the proposal.
- B. Exactly 63% of all voters support the proposal.
- C. The margin of error would stay 5 percentage points if the sample size were much smaller.
- D. The result proves that the proposal will pass.

845. Inference from samples · Hard · MC

In a random sample of 1200 voters, 61% supported a proposal. The margin of error for this estimate is 3 percentage points. Which conclusion is best supported?

- A. The margin of error would stay 3 percentage points if the sample size were much smaller.
- B. The result proves that the proposal will pass.
- C. It is plausible that between 58% and 64% of all voters support the proposal.
- D. Exactly 61% of all voters support the proposal.

846. Evaluating statistical claims · Medium · MC

A school claims that students who join the chess club have higher math grades, so the club causes the higher grades. Why is that claim not well supported?

- A. Observational results are never allowed to include numbers.
- B. Students who choose the club may already differ from students who do not.
- C. The school did not report the margin of error of a poll.
- D. A mean cannot be computed from letter grades.

847. Evaluating statistical claims · Medium · MC

Researchers randomly select classrooms, then randomly assign those classrooms to a new reading program or the current program. The random assignment is useful mainly because it helps

- A. make the margin of error equal to zero.
- B. support a conclusion about the effect of the program.
- C. remove the need to record any scores.
- D. guarantee that every student in the country is represented.

848. Evaluating statistical claims · Medium · MC

To estimate the proportion of all students at a college who live on campus, which method is most appropriate?

- A. Ask only the students in one dormitory.
- B. Assign a random half of the college to move on campus.
- C. Survey students at a different college.
- D. Take a random sample of students from the college and record whether each lives on campus.

849. Evaluating statistical claims · Medium · MC

A poll uses a random sample of city residents. Which change would be expected to reduce the margin of error?

- A. Replace the random sample with volunteers.
- B. *Report the result as a fraction instead of a percent.*
- C. Increase the sample size.
- D. Ask the question in a different order only.

850. Evaluating statistical claims · Medium · MC

A researcher wants to know whether a new homework format causes higher quiz scores for students at one high school. Which design best supports a cause-and-effect conclusion?

- A. Survey a random sample of students about which format they prefer.
- B. Compare this year's quiz scores with last year's after the school changes the format for everyone.
- C. Randomly assign students at the school to the new format or the current format, then compare quiz scores.
- D. Ask volunteers to try the new format and compare them with students who kept the old format.

Geometry and Trigonometry

851. Angles and triangles · Easy · SPR

Two angles of a triangle measure 80° and 35° . What is the measure, in degrees, of the third angle?

852. Angles and triangles · Easy · MC

Two angles of a triangle measure 52° and 72° . What is the measure, in degrees, of the third angle?

A. 124

B. 56

C. 236

D. 90

853. Angles and triangles · Easy · SPR

Two angles of a triangle measure 78° and 52° . What is the measure, in degrees, of the third angle?

854. Angles and triangles · Easy · SPR

Two angles of a triangle measure 31° and 68° . What is the measure, in degrees, of the third angle?

855. Angles and triangles · Easy · MC

Two angles of a triangle measure 57° and 49° . What is the measure, in degrees, of the third angle?

A. 90

B. 254

C. 74

D. 106

856. Angles and triangles · Easy · MC

Two angles of a triangle measure 39° and 57° . What is the measure, in degrees, of the third angle?

A. 90

B. 84

C. 96

D. 264

857. Angles and triangles · Easy · MC

Two angles of a triangle measure 35° and 34° . What is the measure, in degrees, of the third angle?

A. 291

B. 90

C. 111

D. 69

858. Angles and triangles · Easy · MC

Two angles of a triangle measure 61° and 31° . What is the measure, in degrees, of the third angle?

A. 88

B. 90

C. 92

D. 268

859. Angles and triangles · Easy · MC

Two angles of a triangle measure 57° and 39° . What is the measure, in degrees, of the third angle?

- A. 90
- B. 96
- C. 264
- D. 84

860. Angles and triangles · Easy · SPR

Two angles of a triangle measure 71° and 44° . What is the measure, in degrees, of the third angle?

861. Angles and triangles · Easy · SPR

Two angles of a triangle measure 28° and 58° . What is the measure, in degrees, of the third angle?

862. Angles and triangles · Easy · MC

Two angles of a triangle measure 47° and 39° . What is the measure, in degrees, of the third angle?

- A. 90
- B. 274
- C. 86
- D. 94

863. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 80° . What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

- A. 280
- B. 90
- C. 100
- D. 80

864. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 70° . What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

- A. 290
- B. 110
- C. 90
- D. 70

865. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 36° . What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

- A. 324
- B. 90
- C. 36
- D. 144

866. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 76° . What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

- A. 284
- B. 104
- C. 76
- D. 90

867. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 138° . What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

- A. 222
- B. 90
- C. 42
- D. 138

868. Angles and triangles · Medium · SPR

Two parallel lines are cut by a transversal. One interior angle measures 49° . What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

869. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 95° . What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

- A. 90
- B. 95
- C. 85
- D. 265

870. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 41° . What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

- A. 319
- B. 90
- C. 41
- D. 139

871. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 46° . What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

- A. 90
- B. 314
- C. 46
- D. 134

872. Angles and triangles · Medium · MC

Two parallel lines are cut by a transversal. One interior angle measures 37° . What is the measure, in degrees, of the consecutive interior angle on the same side of the transversal?

- A. 90
- B. 37
- C. 323
- D. 143

873. Angles and triangles · Hard · MC

Two parallel lines are cut by a transversal. One angle measures 123° . What is the measure, in degrees, of the alternate interior angle?

- A. 237
- B. 90
- C. 123
- D. 57

874. Angles and triangles · Hard · SPR

Two parallel lines are cut by a transversal. One angle measures 138° . What is the measure, in degrees, of the alternate interior angle?

875. Angles and triangles · Hard · MC

Two parallel lines are cut by a transversal. One angle measures 96° . What is the measure, in degrees, of the alternate interior angle?

- A. 84
- B. 90
- C. 264
- D. 96

876. Angles and triangles · Hard · MC

Two parallel lines are cut by a transversal. One angle measures 145° . What is the measure, in degrees, of the alternate interior angle?

- A. 90
- B. 145
- C. 35
- D. 215

877. Angles and triangles · Hard · MC

Two parallel lines are cut by a transversal. One angle measures 93° . What is the measure, in degrees, of the alternate interior angle?

- A. 87
- B. 90
- C. 93
- D. 267

878. Angles and triangles · Hard · MC

Two parallel lines are cut by a transversal. One angle measures 77° . What is the measure, in degrees, of the alternate interior angle?

- A. 103
- B. 283
- C. 90
- D. 77

879. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 20 and one leg of length 12. What is the length of the other leg?

- A. 8
- B. 32
- C. 20
- D. 16

880. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 35 and one leg of length 28. What is the length of the other leg?

- A. 7
- B. 63
- C. 21
- D. 35

881. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 61 and one leg of length 11. What is the length of the other leg?

- A. 72
- B. 60
- C. 50
- D. 61

882. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 37 and one leg of length 12. What is the length of the other leg?

- A. 25
- B. 35
- C. 37
- D. 49

883. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 20 and one leg of length 16. What is the length of the other leg?

- A. 20
- B. 4
- C. 12
- D. 36

884. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 50 and one leg of length 14. What is the length of the other leg?

- A. 50
- B. 48
- C. 36
- D. 64

885. Right triangles · Easy · MC

A right triangle has a hypotenuse of length 25 and one leg of length 7. What is the length of the other leg?

- A. 25
- B. 18
- C. 24
- D. 32

886. Right triangles · Easy · SPR

A right triangle has a hypotenuse of length 41 and one leg of length 9. What is the length of the other leg?

887. Right triangles · Medium · MC

A right triangle has a hypotenuse of length 34 and one leg of length 30. What is the length of the other leg?

- A. 64
- B. 16
- C. 34
- D. 4

888. Right triangles · Medium · SPR

A right triangle has a hypotenuse of length 13 and one leg of length 5. What is the length of the other leg?

889. Right triangles · Medium · MC

A right triangle has a hypotenuse of length 26 and one leg of length 10. What is the length of the other leg?

- A. 36
- B. 26
- C. 16
- D. 24

890. Right triangles · Medium · MC

A right triangle has a hypotenuse of length 30 and one leg of length 24. What is the length of the other leg?

- A. 6
- B. 54
- C. 18
- D. 30

891. Right triangles · Medium · SPR

A right triangle has a hypotenuse of length 29 and one leg of length 20. What is the length of the other leg?

892. Right triangles · Hard · MC

A right triangle has legs of length 9 and 40. What is the length of the hypotenuse?

- A. 49
- B. 360
- C. 31
- D. 41

893. Right triangles · Hard · SPR

A right triangle has legs of length 24 and 7. What is the length of the hypotenuse?

894. Right triangles · Hard · MC

A right triangle has legs of length 60 and 11. What is the length of the hypotenuse?

- A. 660
- B. 71
- C. 61
- D. 49

895. Right triangles · Hard · MC

A right triangle has legs of length 11 and 60. What is the length of the hypotenuse?

- A. 49
- B. 71
- C. 660
- D. 61

896. Right triangles · Medium · SPR

An isosceles right triangle has a hypotenuse of length $4\sqrt{2}$. What is the length of each leg?

897. Right triangles · Medium · MC

In a 30° - 60° - 90° triangle, the side opposite the 30° angle has length 7. What is the length of the hypotenuse?

- A. 14
- B. 7
- C. 49
- D. 21

898. Right triangles · Medium · MC

In a 30° - 60° - 90° triangle, the side opposite the 30° angle has length 8. What is the length of the hypotenuse?

- A. 24
- B. 8
- C. 16
- D. 64

899. Right triangles · Medium · MC

In a 30° - 60° - 90° triangle, the side opposite the 30° angle has length 2. What is the length of the hypotenuse?

- A. 2
- B. 60
- C. 4
- D. 6

900. Right triangles · Medium · MC

In a 30° - 60° - 90° triangle, the side opposite the 30° angle has length 9. What is the length of the hypotenuse?

- A. 81
- B. 9
- C. 27
- D. 18

901. Right triangles · Medium · SPR

In a 30° - 60° - 90° triangle, the side opposite the 30° angle has length 15. What is the length of the hypotenuse?

902. Right triangles · Medium · MC

In a 30° - 60° - 90° triangle, the side opposite the 30° angle has length 6. What is the length of the hypotenuse?

- A. 36
- B. 6
- C. 18
- D. 12

903. Right triangles · Medium · MC

An isosceles right triangle has a hypotenuse of length $8\sqrt{2}$. What is the length of each leg?

- A. 16
- B. 4
- C. 8
- D. 64

904. Right triangles · Hard · SPR

An isosceles right triangle has a hypotenuse of length $5\sqrt{2}$. What is the length of each leg?

905. Right triangles · Hard · MC

In a 30° - 60° - 90° triangle, the side opposite the 30° angle has length 10. What is the length of the hypotenuse?

- A. 20
- B. 100
- C. 30
- D. 10

906. Right triangles · Hard · MC

An isosceles right triangle has a hypotenuse of length $6\sqrt{2}$. What is the length of each leg?

- A. 36
- B. 6
- C. 3
- D. 12

907. Right triangles · Hard · MC

An isosceles right triangle has a hypotenuse of length $11\sqrt{2}$. What is the length of each leg?

- A. $\frac{11}{2}$
- B. 11
- C. 121
- D. 22

908. Right triangles · Hard · MC

In a 30° - 60° - 90° triangle, the side opposite the 30° angle has length 16. What is the length of the hypotenuse?

- A. 256
- B. 32
- C. 48
- D. 16

909. Right triangles · Hard · MC

An isosceles right triangle has a hypotenuse of length $15\sqrt{2}$. What is the length of each leg?

- A. 15
- B. 225
- C. 30
- D. $\frac{15}{2}$

910. Area and volume · Easy · MC

A rectangle has length 5 and width 7. What is its area?

- A. 24
- B. 35
- C. 12
- D. 37

911. Area and volume · Easy · MC

A rectangle has length 13 and width 7. What is its area?

- A. 40
- B. 93
- C. 20
- D. 91

912. Area and volume · Easy · MC

A rectangle has length 8 and width 7. What is its area?

- A. 15
- B. 30
- C. 58
- D. 56

913. Area and volume · Easy · SPR

A rectangle has length 6 and width 7. What is its area?

914. Area and volume · Easy · MC

A rectangle has length 7 and width 10. What is its area?

A. 70

B. 72

C. 34

D. 17

915. Area and volume · Easy · MC

A rectangle has length 8 and width 9. What is its area?

A. 17

B. 72

C. 34

D. 74

916. Area and volume · Easy · MC

A rectangle has length 5 and width 3. What is its area?

A. 17

B. 16

C. 8

D. 15

917. Area and volume · Easy · MC

A rectangle has length 6 and width 9. What is its area?

A. 56

B. 54

C. 30

D. 15

918. Area and volume · Easy · MC

A rectangle has length 15 and width 6. What is its area?

A. 21

B. 90

C. 42

D. 92

919. Area and volume · Easy · MC

A rectangle has length 6 and width 3. What is its area?

A. 18

B. 20

C. 9

D. 36

920. Area and volume · Medium · SPR

A triangle has a base of length 4 and a corresponding height of 4. What is the area of the triangle?

921. Area and volume · Medium · MC

A triangle has a base of length 15 and a corresponding height of 8. What is the area of the triangle?

- A. 30
- B. 23
- C. 60
- D. 120

922. Area and volume · Medium · MC

A triangle has a base of length 14 and a corresponding height of 10. What is the area of the triangle?

- A. 140
- B. 70
- C. 35
- D. 24

923. Area and volume · Medium · MC

A triangle has a base of length 7 and a corresponding height of 10. What is the area of the triangle?

- A. 35
- B. 70
- C. 17
- D. $\frac{35}{2}$

924. Area and volume · Medium · SPR

A triangle has a base of length 14 and a corresponding height of 12. What is the area of the triangle?

925. Area and volume · Medium · MC

A triangle has a base of length 16 and a corresponding height of 10. What is the area of the triangle?

- A. 160
- B. 26
- C. 80
- D. 40

926. Area and volume · Medium · MC

A triangle has a base of length 15 and a corresponding height of 6. What is the area of the triangle?

- A. 21
- B. 90
- C. $\frac{45}{2}$
- D. 45

927. Area and volume · Medium · MC

A triangle has a base of length 5 and a corresponding height of 6. What is the area of the triangle?

- A. 15
- B. 30
- C. $\frac{15}{2}$
- D. 11

928. Area and volume · Easy · MC

A rectangular prism has length 5, width 8, and height 5. What is the volume of the prism?

- A. 40
- B. 200
- C. 18
- D. 210

929. Area and volume · Easy · MC

A rectangular prism has length 3, width 8, and height 9. What is the volume of the prism?

- A. 20
- B. 216
- C. 246
- D. 24

930. Area and volume · Easy · SPR

A rectangular prism has length 4, width 6, and height 6. What is the volume of the prism?

931. Area and volume · Easy · MC

A rectangular prism has length 8, width 3, and height 9. What is the volume of the prism?

- A. 20
- B. 246
- C. 216
- D. 24

932. Area and volume · Easy · MC

A rectangular prism has length 5, width 4, and height 2. What is the volume of the prism?

- A. 40
- B. 11
- C. 20
- D. 76

933. Area and volume · Easy · MC

A rectangular prism has length 7, width 5, and height 8. What is the volume of the prism?

- A. 35
- B. 262
- C. 280
- D. 20

934. Area and volume · Easy · SPR

A rectangular prism has length 6, width 5, and height 4. What is the volume of the prism?

935. Area and volume · Easy · MC

A rectangular prism has length 3, width 3, and height 8. What is the volume of the prism?

- A. 72
- B. 114
- C. 14
- D. 9

936. Area and volume · Medium · MC

A rectangular prism has length 7, width 4, and height 5. What is the volume of the prism?

- A. 28
- B. 16
- C. 166
- D. 140

937. Area and volume · Medium · MC

A rectangular prism has length 5, width 7, and height 9. What is the volume of the prism?

- A. 21
- B. 35
- C. 315
- D. 286

938. Area and volume · Medium · MC

A rectangular prism has length 3, width 7, and height 6. What is the volume of the prism?

- A. 16
- B. 162
- C. 126
- D. 21

939. Area and volume · Medium · SPR

A rectangular prism has length 9, width 5, and height 2. What is the volume of the prism?

940. Area and volume · Medium · MC

A rectangular prism has length 8, width 5, and height 4. What is the volume of the prism?

- A. 160
- B. 184
- C. 17
- D. 40

941. Area and volume · Medium · SPR

A rectangular prism has length 9, width 7, and height 6. What is the volume of the prism?

942. Area and volume · Hard · MC

A right circular cylinder has radius 6 and height 8. The volume is $k\pi$. What is the value of k ?

- A. 288
- B. 96
- C. 48
- D. 576

943. Area and volume · Hard · MC

A right circular cylinder has radius 4 and height 12. The volume is $k\pi$. What is the value of k ?

- A. 192
- B. 48
- C. 384
- D. 96

944. Area and volume · Hard · SPR

A right circular cylinder has radius 3 and height 11. The volume is $k\pi$. What is the value of k ?

945. Area and volume · Hard · MC

A right circular cylinder has radius 6 and height 9. The volume is $k\pi$. What is the value of k ?

- A. 648
- B. 108
- C. 324
- D. 54

946. Area and volume · Hard · MC

A right circular cylinder has radius 6 and height 10. The volume is $k\pi$. What is the value of k ?

- A. 120
- B. 720
- C. 60
- D. 360

947. Area and volume · Hard · SPR

A right circular cylinder has radius 5 and height 10. The volume is $k\pi$. What is the value of k ?

948. Area and volume · Hard · SPR

A right circular cylinder has radius 4 and height 4. The volume is $k\pi$. What is the value of k ?

949. Area and volume · Hard · MC

A right circular cylinder has radius 2 and height 12. The volume is $k\pi$. What is the value of k ?

- A. 16
- B. 96
- C. 48
- D. 24

950. Circles · Easy · MC

A circle has radius 8. The circumference is $k\pi$. What is the value of k ?

- A. 16
- B. 64
- C. 8
- D. 128

951. Circles · Easy · MC

A circle has radius 7. The circumference is $k\pi$. What is the value of k ?

- A. 7
- B. 14
- C. 98
- D. 49

952. Circles · Easy · MC

A circle has radius 4. The circumference is $k\pi$. What is the value of k ?

- A. 32
- B. 16
- C. 8
- D. 4

953. Circles · Easy · MC

A circle has radius 5. The circumference is $k\pi$. What is the value of k ?

- A. 10
- B. 50
- C. 5
- D. 25

954. Circles · Easy · SPR

A circle has radius 6. The circumference is $k\pi$. What is the value of k ?

955. Circles · Easy · MC

A circle has radius 12. The circumference is $k\pi$. What is the value of k ?

- A. 144
- B. 288
- C. 24
- D. 12

956. Circles · Medium · SPR

What is the radius of the circle with equation $(x - 2)^2 + (y - 7)^2 = 25$?

957. Circles · Medium · MC

What is the radius of the circle with equation $(x - 3)^2 + (y + 5)^2 = 49$?

- A. 7
- B. 14
- C. 3
- D. 49

958. Circles · Medium · MC

What is the radius of the circle with equation $(x + 2)^2 + (y - 5)^2 = 36$?

- A. 6
- B. -2
- C. 12
- D. 36

959. Circles · Medium · MC

What is the radius of the circle with equation $(x + 1)^2 + (y + 3)^2 = 4$?

- A. 4
- B. -1
- C. 2
- D. -3

960. Circles · Medium · MC

What is the radius of the circle with equation $(x + 3)^2 + (y + 6)^2 = 49$?

- A. 14
- B. 7
- C. -3
- D. 49

961. Circles · Medium · MC

What is the radius of the circle with equation $(x + 5)^2 + (y + 3)^2 = 16$?

- A. 16
- B. -5
- C. 4
- D. 8

962. Circles · Medium · SPR

What is the radius of the circle with equation $(x + 2)^2 + (y - 1)^2 = 36$?

963. Circles · Medium · MC

What is the radius of the circle with equation $(x - 4)^2 + (y - 3)^2 = 49$?

- A. 49
- B. 4
- C. 7
- D. 14

964. Circles · Hard · MC

What is the radius of the circle with equation $x^2 - 6x + y^2 + 8y = -9$?

- A. 3
- B. 16
- C. 4
- D. 8

965. Circles · Hard · MC

What is the radius of the circle with equation $x^2 - 8x + y^2 + 6y = -9$?

- A. 4
- B. 3
- C. 16
- D. 8

966. Circles · Hard · MC

What is the radius of the circle with equation $x^2 - 8x + y^2 + 8y = -23$?

- A. 3
- B. 6
- C. 4
- D. 9

967. Circles · Hard · MC

What is the radius of the circle with equation $x^2 - 4x + y^2 + 8y = -11$?

- A. 3
- B. 2
- C. 9
- D. 4

968. Circles · Hard · MC

What is the radius of the circle with equation $x^2 - 6x + y^2 + 4y = 23$?

- A. 6
- B. 2
- C. 3
- D. 36

969. Circles · Hard · MC

What is the radius of the circle with equation $x^2 - 10x + y^2 + 8y = -32$?

- A. 5
- B. 9
- C. 3
- D. 4

970. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 4 and 16. Another side of the smaller triangle measures 10. What is the length of the corresponding side of the larger triangle?

- A. $\frac{5}{2}$
- B. 44
- C. 14
- D. 40

971. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 5 and 10. Another side of the smaller triangle measures 7. What is the length of the corresponding side of the larger triangle?

- A. 14
- B. 35
- C. 19
- D. 9

972. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 6 and 12. Another side of the smaller triangle measures 12. What is the length of the corresponding side of the larger triangle?

- A. 24
- B. 14
- C. 72
- D. 30

973. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 4 and 12. Another side of the smaller triangle measures 10. What is the length of the corresponding side of the larger triangle?

- A. 13
- B. 30
- C. 40
- D. 34

974. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 7 and 14. Another side of the smaller triangle measures 10. What is the length of the corresponding side of the larger triangle?

- A. 70
- B. 20
- C. 12
- D. 27

975. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 6 and 18. Another side of the smaller triangle measures 9. What is the length of the corresponding side of the larger triangle?

- A. 33
- B. 27
- C. 12
- D. 54

976. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 7 and 28. Another side of the smaller triangle measures 10. What is the length of the corresponding side of the larger triangle?

- A. 70
- B. 14
- C. 47
- D. 40

977. Angles and triangles · Medium · MC

Two triangles are similar. A pair of corresponding sides measure 8 and 32. Another side of the smaller triangle measures 12. What is the length of the corresponding side of the larger triangle?

- A. 48
- B. 16
- C. 56
- D. 96

978. Angles and triangles · Hard · MC

Two triangles are similar. A pair of corresponding sides measure 6 and 24. Another side of the smaller triangle measures 8. What is the length of the corresponding side of the larger triangle?

- A. 12
- B. 38
- C. 32
- D. 48

979. Angles and triangles · Hard · SPR

Two triangles are similar. A pair of corresponding sides measure 5 and 15. Another side of the smaller triangle measures 7. What is the length of the corresponding side of the larger triangle?

980. Angles and triangles · Hard · MC

Two triangles are similar. A pair of corresponding sides measure 6 and 18. Another side of the smaller triangle measures 8. What is the length of the corresponding side of the larger triangle?

- A. 11
- B. 48
- C. 24
- D. 30

981. Angles and triangles · Hard · SPR

Two triangles are similar. A pair of corresponding sides measure 8 and 16. Another side of the smaller triangle measures 11. What is the length of the corresponding side of the larger triangle?

982. Angles and triangles · Hard · MC

Two triangles are similar. A pair of corresponding sides measure 4 and 12. Another side of the smaller triangle measures 9. What is the length of the corresponding side of the larger triangle?

- A. 12
- B. 27
- C. 36
- D. 31

983. Angles and triangles · Hard · SPR

Two triangles are similar. A pair of corresponding sides measure 8 and 24. Another side of the smaller triangle measures 11. What is the length of the corresponding side of the larger triangle?

984. Trigonometry · Easy · MC

In a right triangle, the side opposite angle A has length 8, the side adjacent to angle A has length 15, and the hypotenuse has length 17. What is the value of $\tan A$?

- A. $\frac{17}{8}$
- B. $\frac{15}{17}$
- C. $\frac{15}{8}$
- D. $\frac{8}{15}$

985. Trigonometry · Easy · SPR

In a right triangle, the side opposite angle A has length 7, the side adjacent to angle A has length 24, and the hypotenuse has length 25. What is the value of $\sin A$?

986. Trigonometry · Easy · MC

In a right triangle, the side opposite angle A has length 5, the side adjacent to angle A has length 12, and the hypotenuse has length 13. What is the value of $\cos A$?

- A. $\frac{5}{12}$
- B. $\frac{12}{13}$
- C. $\frac{12}{5}$
- D. $\frac{13}{5}$

987. Trigonometry · Easy · MC

In a right triangle, the side opposite angle A has length 9, the side adjacent to angle A has length 12, and the hypotenuse has length 15. What is the value of $\cos A$?

- A. $\frac{3}{4}$
- B. $\frac{4}{3}$
- C. $\frac{5}{3}$
- D. $\frac{4}{5}$

988. Trigonometry · Easy · SPR

In a right triangle, the side opposite angle A has length 12, the side adjacent to angle A has length 35, and the hypotenuse has length 37. What is the value of $\tan A$?

989. Trigonometry · Easy · MC

In a right triangle, the side opposite angle A has length 20, the side adjacent to angle A has length 21, and the hypotenuse has length 29. What is the value of $\cos A$?

- A. $\frac{21}{20}$
- B. $\frac{20}{21}$
- C. $\frac{21}{29}$
- D. $\frac{29}{20}$

990. Trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 6, the side adjacent to angle A has length 8, and the hypotenuse has length 10. What is the value of $\tan A$?

- A. $\frac{4}{3}$
- B. $\frac{5}{3}$
- C. $\frac{3}{4}$
- D. $\frac{4}{5}$

991. Trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 12, the side adjacent to angle A has length 35, and the hypotenuse has length 37. What is the value of $\sin A$?

- A. $\frac{37}{12}$
- B. $\frac{12}{35}$
- C. $\frac{12}{37}$
- D. $\frac{35}{37}$

992. Trigonometry · Medium · SPR

In a right triangle, the side opposite angle A has length 6, the side adjacent to angle A has length 8, and the hypotenuse has length 10. What is the value of $\cos A$?

993. Trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 12, the side adjacent to angle A has length 16, and the hypotenuse has length 20. What is the value of $\cos A$?

- A. $\frac{4}{5}$
- B. $\frac{5}{3}$
- C. $\frac{4}{3}$
- D. $\frac{3}{4}$

994. Trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 7, the side adjacent to angle A has length 24, and the hypotenuse has length 25. What is the value of $\cos A$?

- A. $\frac{7}{24}$
- B. $\frac{24}{25}$
- C. $\frac{25}{7}$
- D. $\frac{24}{7}$

995. Trigonometry · Medium · MC

In a right triangle, the side opposite angle A has length 12, the side adjacent to angle A has length 35, and the hypotenuse has length 37. What is the value of $\cos A$?

- A. $\frac{35}{37}$
- B. $\frac{12}{35}$
- C. $\frac{35}{12}$
- D. $\frac{37}{12}$

996. Trigonometry · Hard · MC

In a right triangle, the side opposite angle A has length 9, the side adjacent to angle A has length 12, and the hypotenuse has length 15. What is the value of $\sin A$?

- A. $\frac{5}{3}$
- B. $\frac{4}{5}$
- C. $\frac{3}{5}$
- D. $\frac{3}{4}$

997. Trigonometry · Hard · MC

In a right triangle, the side opposite angle A has length 12, the side adjacent to angle A has length 16, and the hypotenuse has length 20. What is the value of $\tan A$?

- A. $\frac{4}{3}$
- B. $\frac{4}{5}$
- C. $\frac{3}{4}$
- D. $\frac{5}{3}$

998. Trigonometry · Hard · SPR

In a right triangle, the side opposite angle A has length 5, the side adjacent to angle A has length 12, and the hypotenuse has length 13. What is the value of $\sin A$?

999. Trigonometry · Hard · MC

In a right triangle, the side opposite angle A has length 6, the side adjacent to angle A has length 8, and the hypotenuse has length 10. What is the value of $\sin A$?

- A. $\frac{5}{3}$
- B. $\frac{3}{5}$
- C. $\frac{4}{5}$
- D. $\frac{3}{4}$

1000. Trigonometry · Hard · MC

In a right triangle, the side opposite angle A has length 3, the side adjacent to angle A has length 4, and the hypotenuse has length 5. What is the value of $\sin A$?

- A. $\frac{3}{5}$
- B. $\frac{3}{4}$
- C. $\frac{4}{5}$
- D. $\frac{5}{3}$

Answer key

Student-produced responses are printed as the number to enter. Multiple-choice answers are the letter.

1. A	52. B	103. D	154. D	205. D
2. A	53. -4	104. C	155. C	206. C
3. 2	54. 7	105. D	156. D	207. D
4. D	55. D	106. A	157. A	208. D
5. 8	56. B	107. D	158. C	209. -1/3
6. B	57. B	108. C	159. B	210. -1/2
7. D	58. A	109. A	160. C	211. -3
8. B	59. -4	110. A	161. B	212. B
9. A	60. -3	111. B	162. 3	213. -1/3
10. A	61. A	112. D	163. B	214. A
11. B	62. B	113. A	164. 3	215. D
12. 6	63. 10	114. A	165. D	216. C
13. B	64. D	115. B	166. 1/2	217. D
14. C	65. C	116. B	167. A	218. A
15. 5	66. -1	117. D	168. A	219. C
16. A	67. A	118. A	169. D	220. B
17. 6	68. B	119. A	170. C	221. D
18. C	69. D	120. A	171. 3	222. C
19. B	70. A	121. C	172. D	223. D
20. 3	71. B	122. C	173. C	224. B
21. 10	72. D	123. C	174. 2/3	225. D
22. B	73. A	124. B	175. B	226. C
23. 3	74. A	125. C	176. B	227. D
24. 7	75. B	126. B	177. C	228. C
25. D	76. B	127. B	178. 3	229. A
26. A	77. B	128. C	179. 1	230. B
27. C	78. C	129. C	180. C	231. B
28. 11	79. D	130. C	181. B	232. B
29. B	80. A	131. D	182. 3	233. B
30. 3	81. C	132. D	183. B	234. A
31. A	82. C	133. A	184. A	235. B
32. A	83. -3	134. A	185. C	236. B
33. C	84. A	135. D	186. B	237. A
34. D	85. 3	136. C	187. A	238. A
35. B	86. B	137. D	188. 16	239. D
36. A	87. C	138. A	189. C	240. C
37. A	88. D	139. D	190. A	241. B
38. C	89. D	140. B	191. C	242. A
39. D	90. B	141. B	192. D	243. D
40. D	91. C	142. D	193. D	244. A
41. D	92. B	143. A	194. C	245. A
42. 4	93. 8	144. C	195. D	246. 2
43. B	94. C	145. A	196. 9	247. -3
44. A	95. 8	146. A	197. C	248. B
45. 9	96. C	147. D	198. 8	249. A
46. A	97. B	148. D	199. D	250. B
47. 8	98. D	149. C	200. 4	251. C
48. 4	99. A	150. A	201. B	252. D
49. 5	100. -4	151. B	202. D	253. A
50. D	101. B	152. B	203. B	254. C
51. 7	102. A	153. A	204. A	255. A

256. C	311. C	366. D	421. -4	476. 44
257. B	312. 5	367. C	422. C	477. 324
258. C	313. 15	368. B	423. D	478. C
259. D	314. 5	369. B	424. 9	479. 640
260. D	315. B	370. B	425. A	480. 63
261. C	316. A	371. C	426. 9	481. B
262. C	317. 10	372. B	427. D	482. C
263. B	318. C	373. B	428. C	483. D
264. -1	319. C	374. A	429. C	484. B
265. B	320. A	375. A	430. A	485. 16
266. D	321. D	376. B	431. A	486. C
267. B	322. D	377. C	432. 4	487. D
268. 5	323. D	378. A	433. -2	488. D
269. A	324. D	379. A	434. 3	489. A
270. B	325. 4	380. B	435. C	490. 1000
271. D	326. 7	381. D	436. A	491. 800
272. A	327. C	382. C	437. 8	492. D
273. D	328. B	383. D	438. 7	493. 640
274. B	329. B	384. B	439. D	494. D
275. D	330. D	385. B	440. D	495. 1600
276. 8	331. 4	386. D	441. B	496. D
277. 7	332. D	387. C	442. B	497. A
278. 7	333. 14	388. B	443. 8	498. B
279. D	334. 11	389. D	444. C	499. D
280. A	335. C	390. A	445. D	500. C
281. C	336. B	391. D	446. D	501. 18
282. B	337. B	392. A	447. A	502. C
283. C	338. A	393. C	448. A	503. 12
284. 1	339. A	394. B	449. A	504. 3
285. 1	340. 13	395. B	450. C	505. C
286. D	341. D	396. A	451. B	506. D
287. 7	342. B	397. B	452. C	507. D
288. A	343. A	398. D	453. D	508. D
289. C	344. B	399. B	454. B	509. D
290. A	345. 11	400. C	455. A	510. C
291. A	346. C	401. -3	456. A	511. C
292. C	347. 12	402. 6	457. D	512. C
293. 9	348. C	403. D	458. D	513. A
294. A	349. C	404. C	459. 7	514. A
295. 7	350. 25	405. B	460. B	515. B
296. A	351. A	406. B	461. 9	516. D
297. B	352. B	407. 3	462. D	517. C
298. A	353. A	408. 11	463. B	518. B
299. 6	354. C	409. A	464. -3	519. B
300. C	355. A	410. C	465. A	520. D
301. D	356. A	411. B	466. A	521. D
302. D	357. C	412. 7	467. C	522. A
303. C	358. D	413. C	468. A	523. C
304. C	359. A	414. A	469. C	524. B
305. B	360. A	415. C	470. C	525. D
306. A	361. A	416. D	471. A	526. D
307. C	362. C	417. C	472. 2048	527. B
308. A	363. C	418. 1	473. 64	528. C
309. A	364. B	419. C	474. C	529. D
310. B	365. C	420. A	475. A	530. A
				531. A
				532. A

533. D	588. 8	643. C	698. B	753. 176
534. C	589. A	644. A	699. D	754. D
535. B	590. 5	645. D	700. B	755. D
536. -9	591. B	646. D	701. D	756. 147
537. 19	592. A	647. D	702. C	757. A
538. A	593. D	648. C	703. D	758. D
539. 2	594. A	649. C	704. A	759. B
540. 3	595. A	650. C	705. A	760. A
541. A	596. B	651. B	706. 36	761. C
542. C	597. 76	652. D	707. C	762. B
543. 17	598. D	653. 3	708. D	763. C
544. 0	599. D	654. 6	709. A	764. D
545. D	600. A	655. A	710. 44	765. D
546. 16	601. A	656. B	711. 36	766. B
547. A	602. B	657. C	712. 35	767. A
548. 4	603. A	658. 81	713. C	768. 15
549. C	604. A	659. 121	714. 30	769. 11
550. B	605. B	660. A	715. 54	770. A
551. B	606. A	661. D	716. 36	771. C
552. C	607. C	662. A	717. C	772. D
553. D	608. B	663. B	718. D	773. A
554. 31	609. 30	664. C	719. 50	774. A
555. 42	610. B	665. D	720. 32	775. D
556. B	611. 5	666. A	721. D	776. D
557. C	612. B	667. D	722. 24	777. A
558. D	613. 14	668. B	723. 64	778. B
559. A	614. A	669. A	724. C	779. D
560. A	615. 44	670. C	725. A	780. D
561. C	616. -1	671. C	726. A	781. A
562. B	617. 0	672. A	727. C	782. C
563. D	618. 6	673. A	728. D	783. B
564. A	619. C	674. B	729. 150	784. C
565. A	620. 4	675. C	730. D	785. D
566. A	621. A	676. D	731. 30	786. A
567. C	622. B	677. C	732. C	787. D
568. C	623. 5	678. D	733. C	788. A
569. C	624. D	679. A	734. D	789. A
570. B	625. C	680. A	735. B	790. B
571. A	626. D	681. A	736. 160	791. B
572. B	627. D	682. A	737. A	792. C
573. A	628. D	683. A	738. D	793. A
574. 10	629. D	684. A	739. C	794. 7/13
575. B	630. D	685. C	740. B	795. C
576. B	631. 2	686. A	741. A	796. D
577. D	632. C	687. C	742. 102	797. C
578. C	633. 8	688. D	743. D	798. 7/12
579. D	634. B	689. B	744. A	799. 3/13
580. B	635. D	690. B	745. 54	800. D
581. D	636. 2	691. C	746. A	801. D
582. C	637. D	692. C	747. 165	802. B
583. 7	638. A	693. C	748. 44	803. D
584. C	639. D	694. D	749. C	804. 1/2
585. C	640. 6	695. D	750. C	805. C
586. B	641. 1	696. D	751. D	806. 3/5
587. B	642. C	697. A	752. C	807. D
				808. D
				809. C

810. A	849. C	888. 12	927. A	966. A
811. B	850. C	889. D	928. B	967. A
812. B	851. 65	890. C	929. B	968. A
813. $\frac{3}{5}$	852. B	891. 21	930. 144	969. C
814. A	853. 50	892. D	931. C	970. D
815. B	854. 81	893. 25	932. A	971. A
816. A	855. C	894. C	933. C	972. A
817. $\frac{4}{7}$	856. B	895. D	934. 120	973. B
818. C	857. C	896. 4	935. A	974. B
819. $\frac{17}{26}$	858. A	897. A	936. D	975. B
820. D	859. D	898. C	937. C	976. D
821. D	860. 65	899. C	938. C	977. A
822. C	861. 94	900. D	939. 90	978. C
823. C	862. D	901. 30	940. A	979. 21
824. B	863. C	902. D	941. 378	980. C
825. B	864. B	903. C	942. A	981. 22
826. C	865. D	904. 5	943. A	982. B
827. D	866. B	905. A	944. 99	983. 33
828. B	867. C	906. B	945. C	984. D
829. -11	868. 131	907. B	946. D	985. $\frac{7}{25}$
830. B	869. C	908. B	947. 250	986. B
831. C	870. D	909. A	948. 64	987. D
832. D	871. D	910. B	949. C	988. $\frac{12}{35}$
833. A	872. D	911. D	950. A	989. C
834. A	873. C	912. D	951. B	990. C
835. 18	874. 138	913. 42	952. C	991. C
836. D	875. D	914. A	953. A	992. $\frac{4}{5}$
837. A	876. B	915. B	954. 12	993. A
838. C	877. C	916. D	955. C	994. B
839. B	878. D	917. B	956. 5	995. A
840. C	879. D	918. B	957. A	996. C
841. C	880. C	919. A	958. A	997. C
842. D	881. B	920. 8	959. C	998. $\frac{5}{13}$
843. A	882. B	921. C	960. B	999. B
844. A	883. C	922. B	961. C	1000. A
845. C	884. B	923. A	962. 6	
846. B	885. C	924. 84	963. C	
847. B	886. 40	925. C	964. C	
848. D	887. B	926. D	965. A	